

THE INTERNATIONAL Calculator Collector

Spring 1993

Issue No. 1



like Cat Tech circa 1967

Photo Courtesy Texas Instruments

The Beginning

If you're past your mid-30s, you probably remember your first simple hand-held calculator costing over \$50 (in early 1970's dollars). Depending how much older you are, your first could have been upwards to \$400. And we're just talking the basic four functions here — addition, subtraction, multiplication, and division. Percentage and memory features were extra (if they were even available at that point in time)

Company Profile:



Who can forget the "Bowmar Brain" series of calculators from the early '70s?

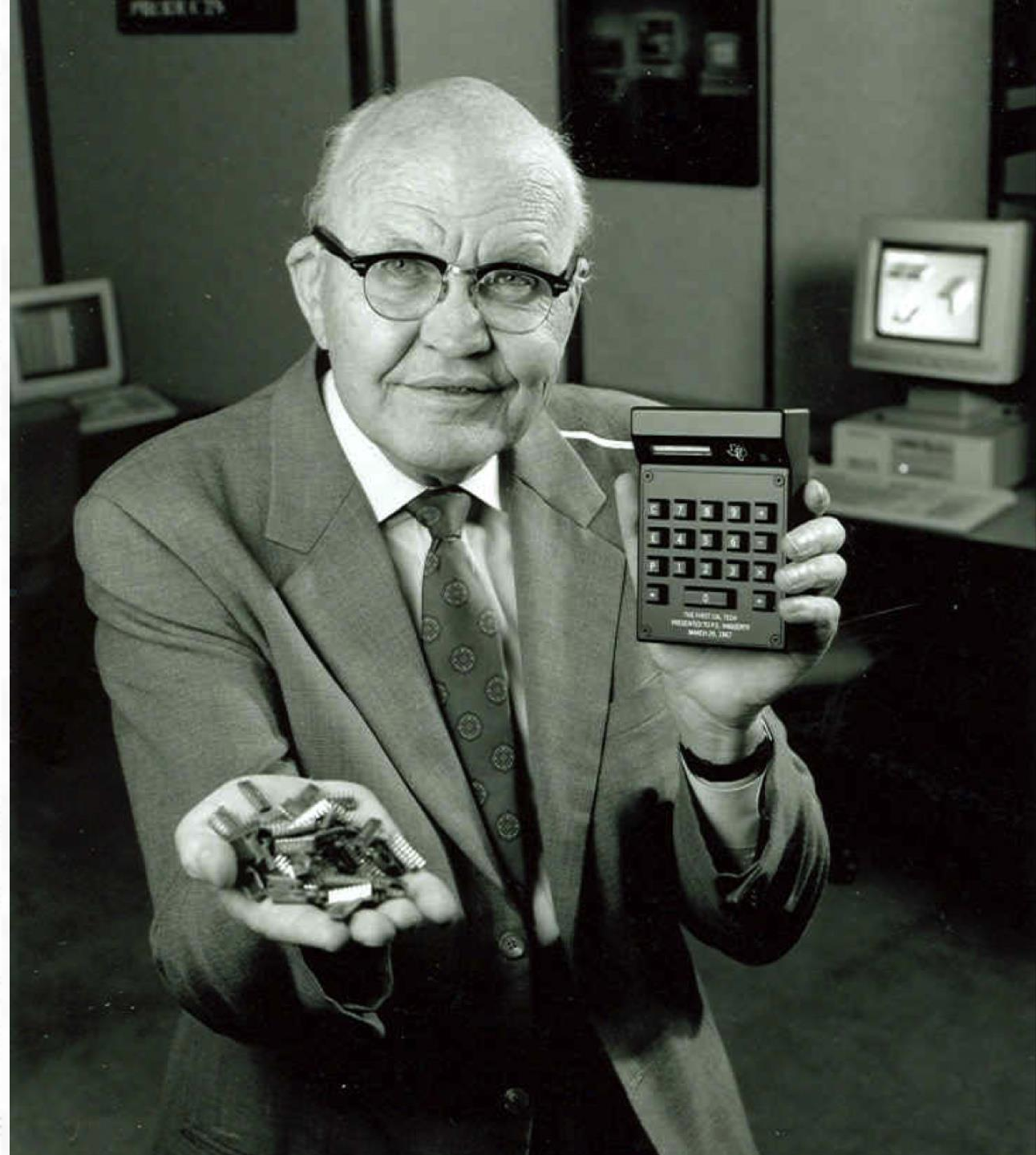
Bowmar was the first American company that made and sold their own line of portable electronic machines.

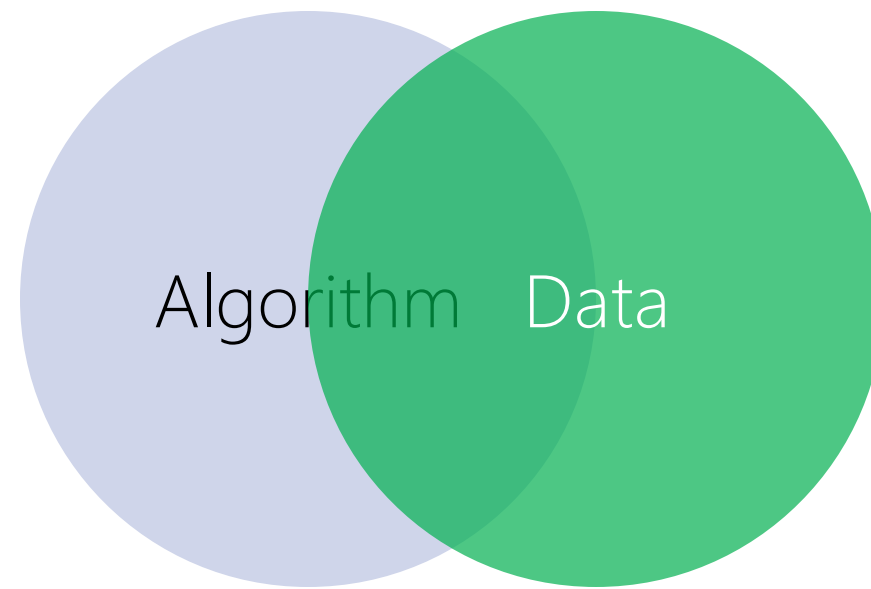
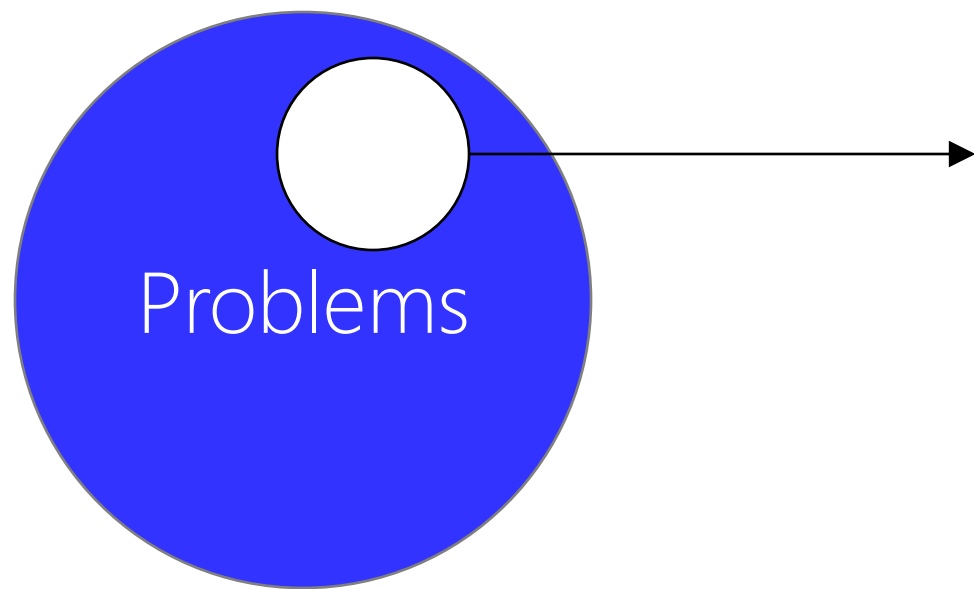
The story starts around 1970 when Bowmar, then a manufacturer of Light Emitting Diodes (LEDs), tried to sell their numeric display product to Japanese manufacturers for use in their electronic products.

Bowmar wasn't too successful. The Japanese were using a fluorescent style display that was cheaper and had a few design features the manufacturers liked better.

So, president Ed White, a consummate entrepreneur, and his staff came up with an even better idea — make the whole electronic calculator themselves.

Up to now, most of the so-called "portable" calculators





Design a Computer System

John von Neumann

([/vɒn 'nɔɪmən/](#))

1903 –1957

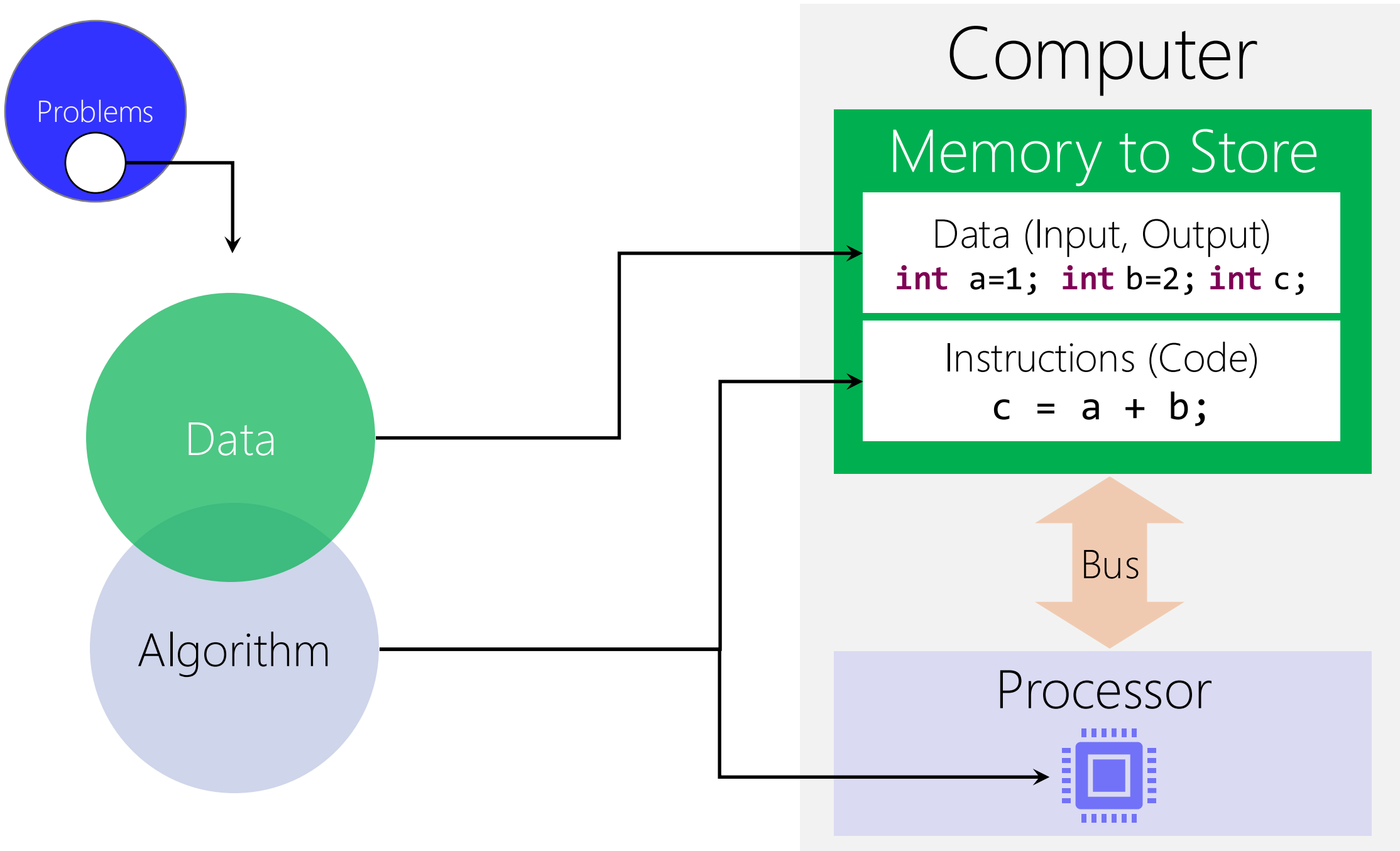
Mathematician, Physicist, Computer Scientist, Engineer

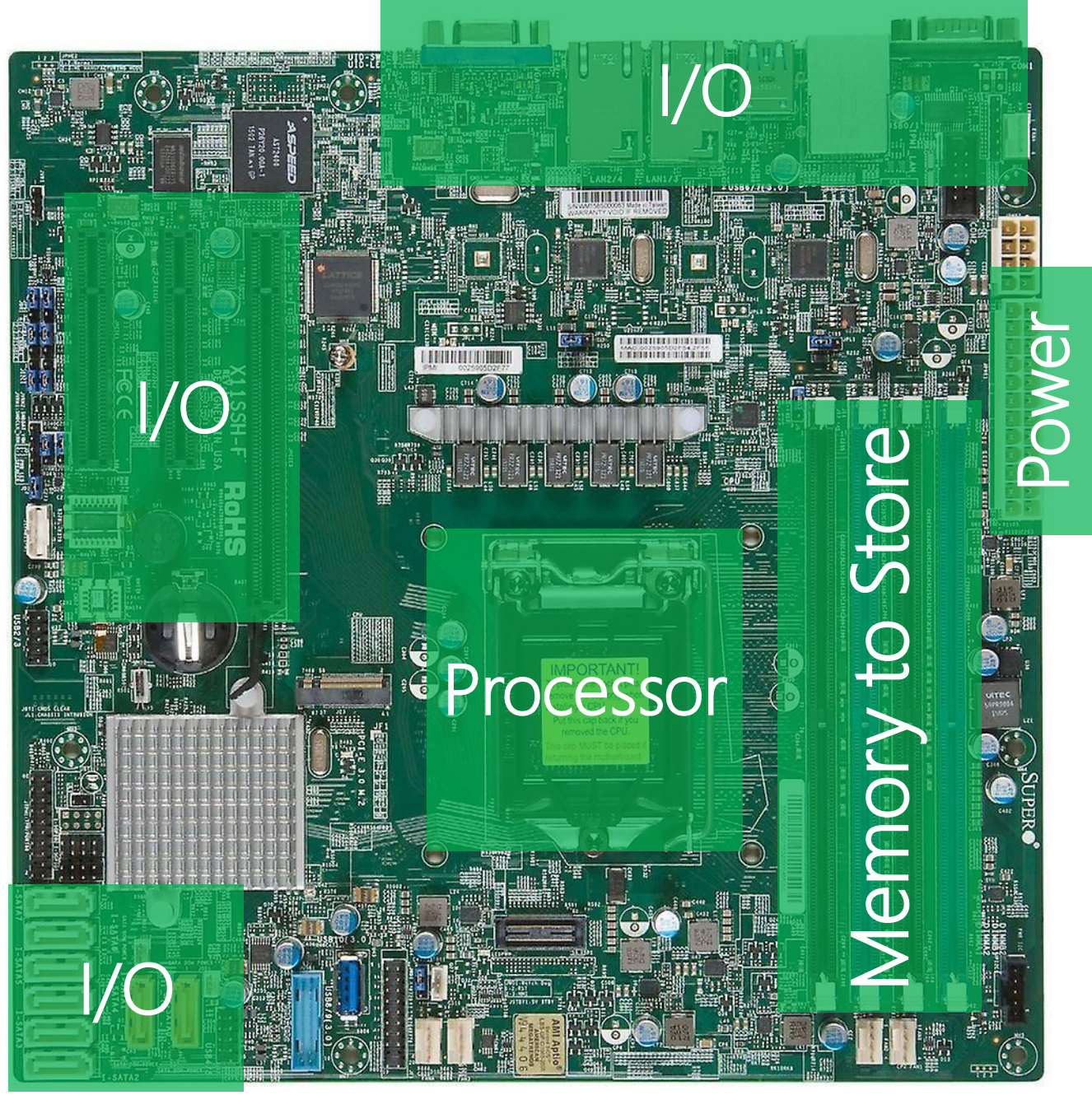
Polymath

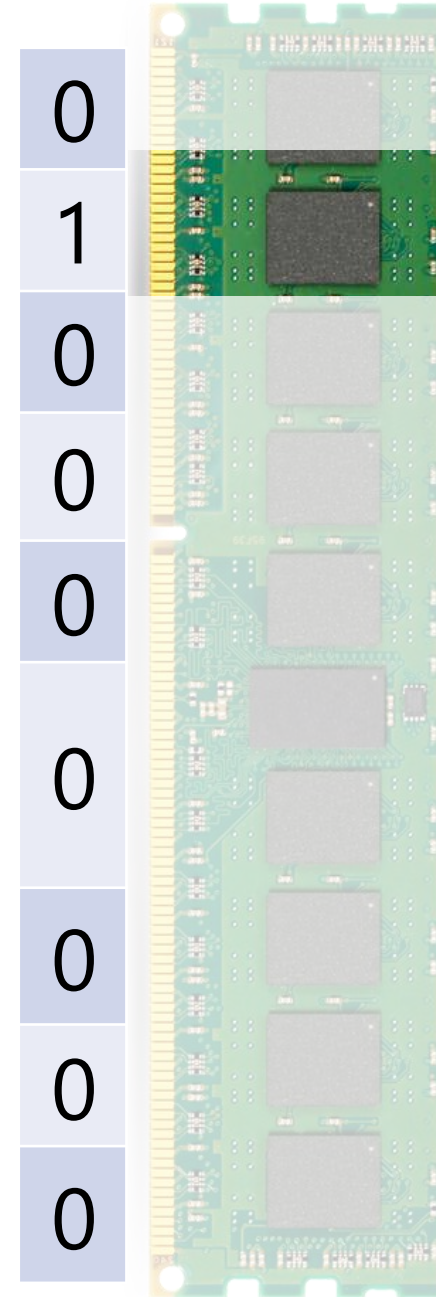
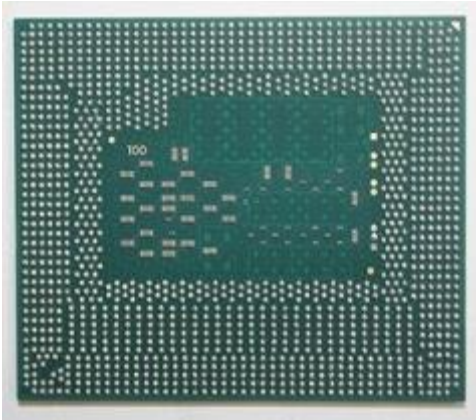
He integrated pure and applied sciences. He made major contributions to many fields, including:

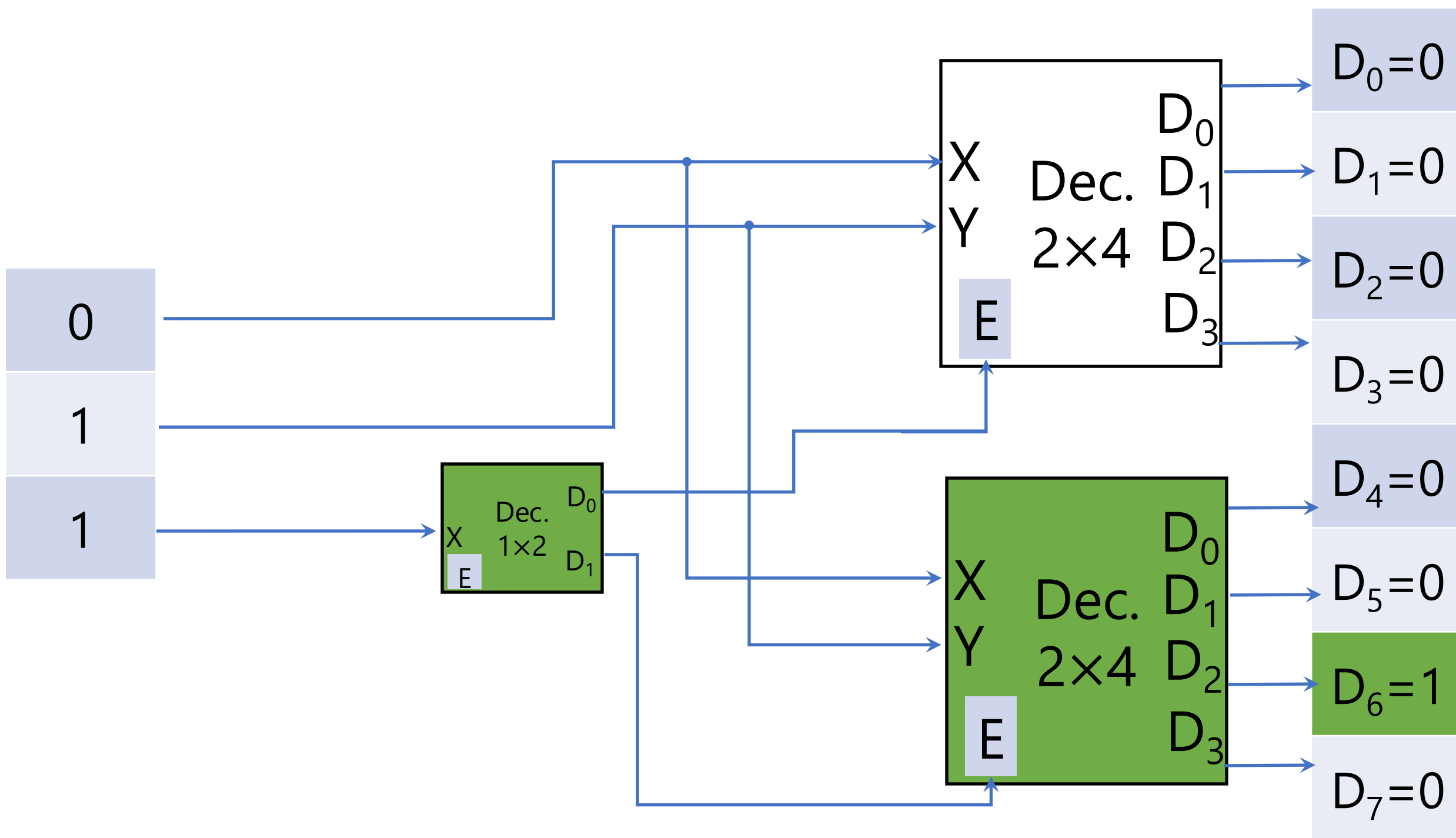
- Mathematics
- Physics
- Economics (game theory)
- Computing
- Statistics









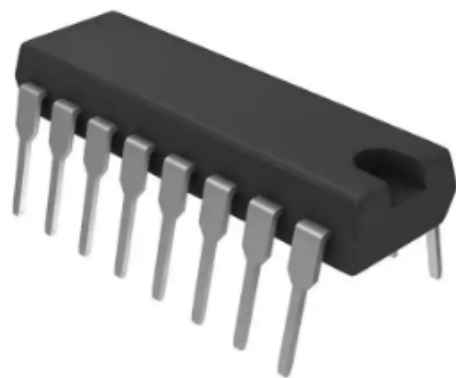




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SN74LS138N

[Datasheet](#)

Digi-Key Part Number	296-1639-5-ND
Manufacturer	Texas Instruments
Manufacturer Product Number	SN74LS138N
Supplier	Texas Instruments
Description	IC 3-8 LINE DECODER/DEMUX 16-DIP

Manufacturer Standard Lead Time 6 Weeks

Decoder/Demultiplexer 1 x 3:8
16-PDIP[Customer Reference](#)

Price and Procurement

4,043 In Stock
Can ship immediately

QUANTITY

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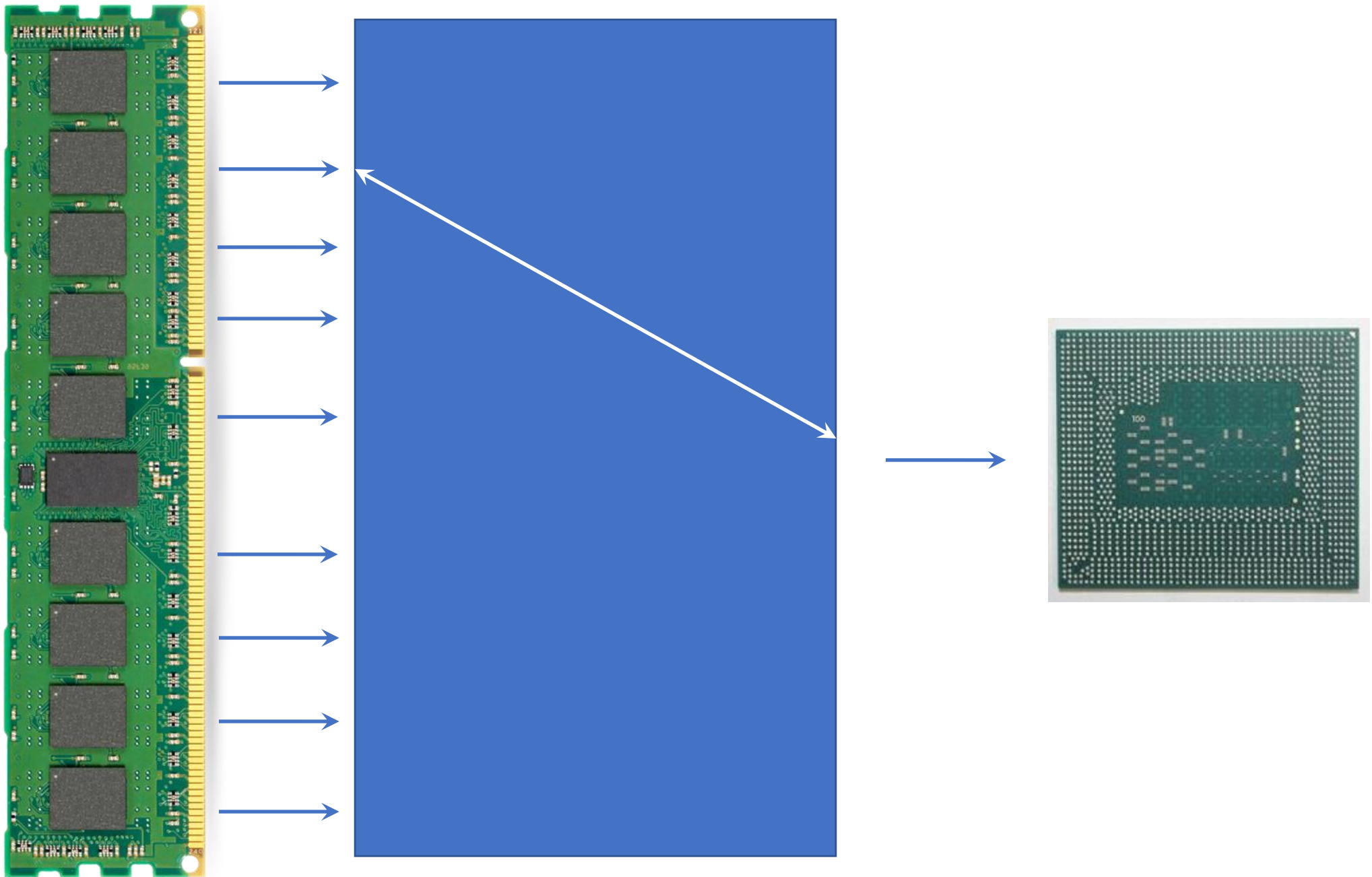
QTY	UNIT PRICE	EXT PRICE
1	\$1.27000	\$1.27
10	\$1.12000	\$11.20
25	\$1.05280	\$26.32
100	\$0.85920	\$85.92

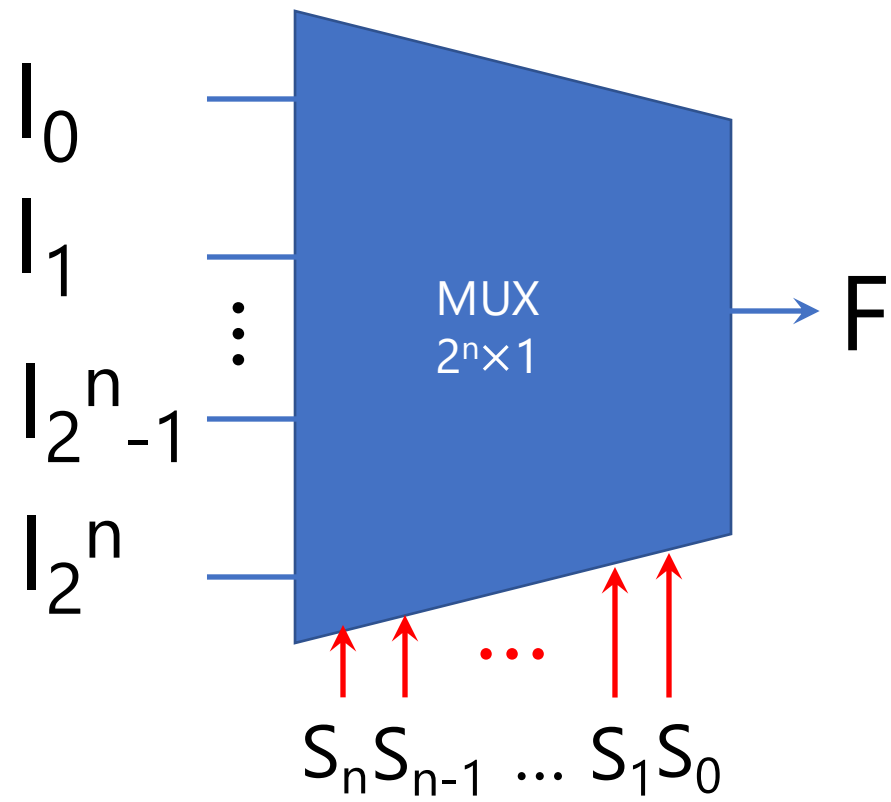
Media & Downloads

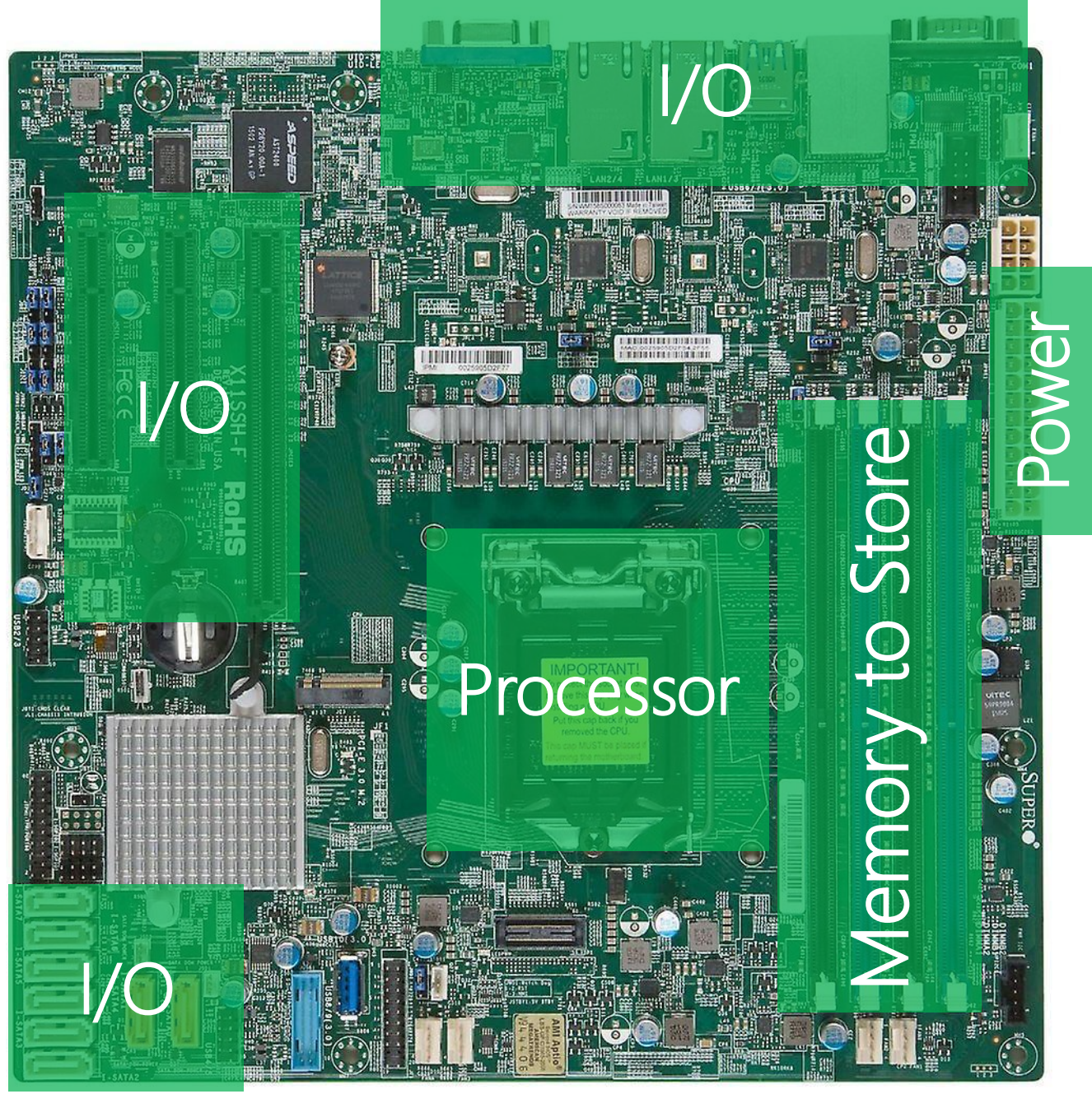
RESOURCE TYPE	LINK
Datasheets	SN54LS138, SN54S138, SN74LS138, SN74S138A
Featured Product	Logic Solutions Analog Solutions
PCN Design/Specification	Material Set 30/Mar/2017
EDA / CAD Models	SN74LS138N by SnapEDA SN74LS138N by Ultra Librarian

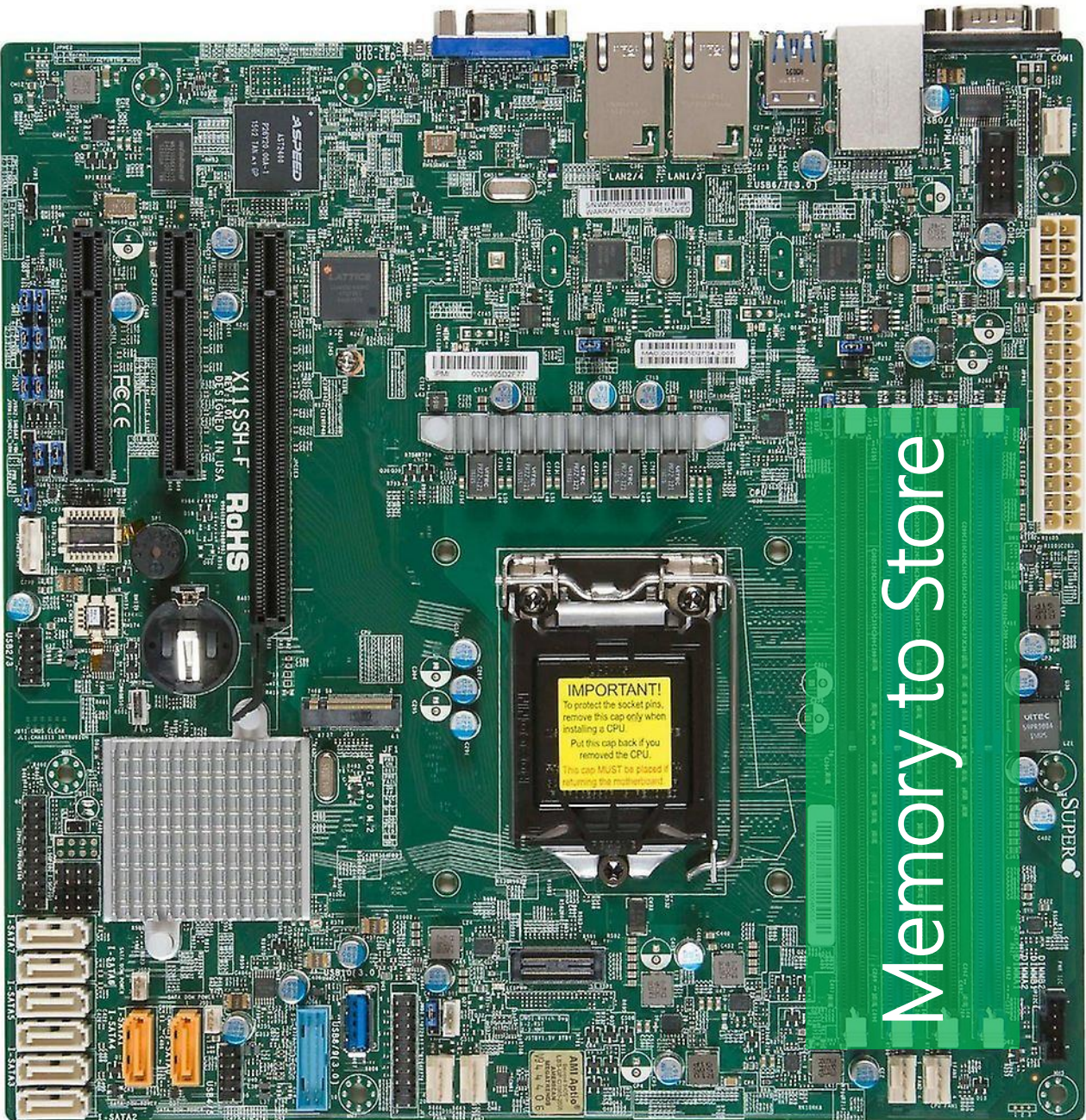
Multiplexer

Shortened to MUX or MPX









Memory to Store

Design Memory

To Store Data + Code

Design Memory

Data → Numbers

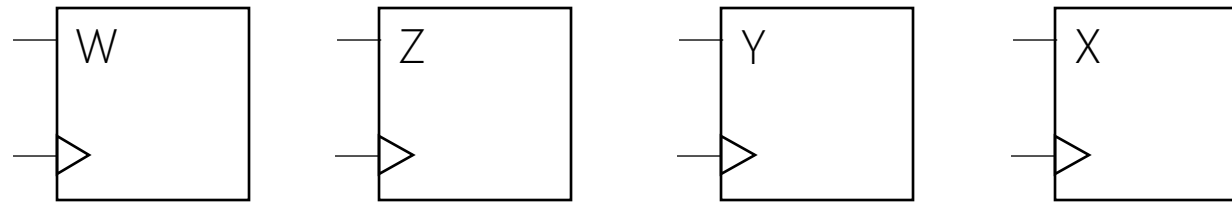
Design Memory

Each Number $\in [\text{min}, \text{max}] \rightarrow \text{Binary System}$

Design Memory

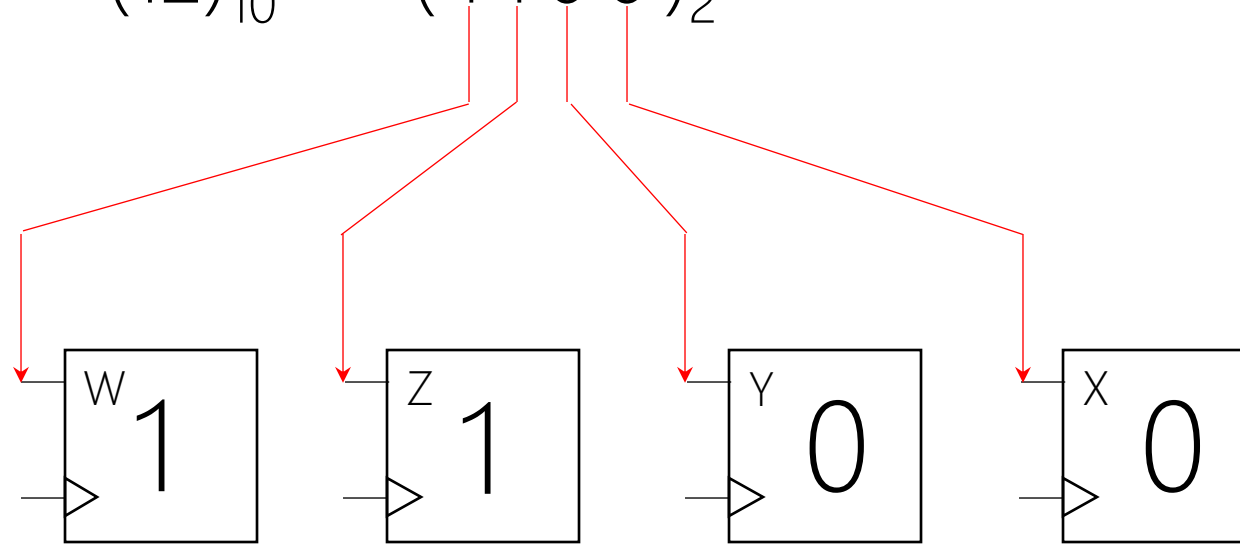
Each Number $\in [0, 2^n) \rightarrow n$ Cells

Number System's Position $\leftrightarrow$ Cells



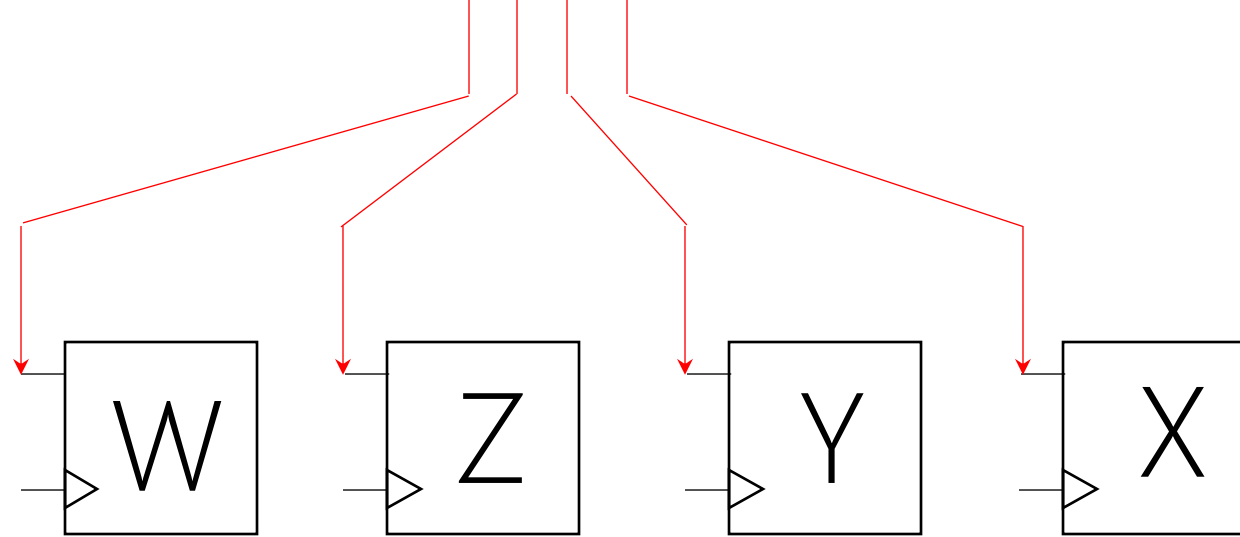
$n=4 \rightarrow$ Each Number $\in [0, 2^4=16) \rightarrow 4$ Cells

$$(12)_{10} = (1100)_2$$



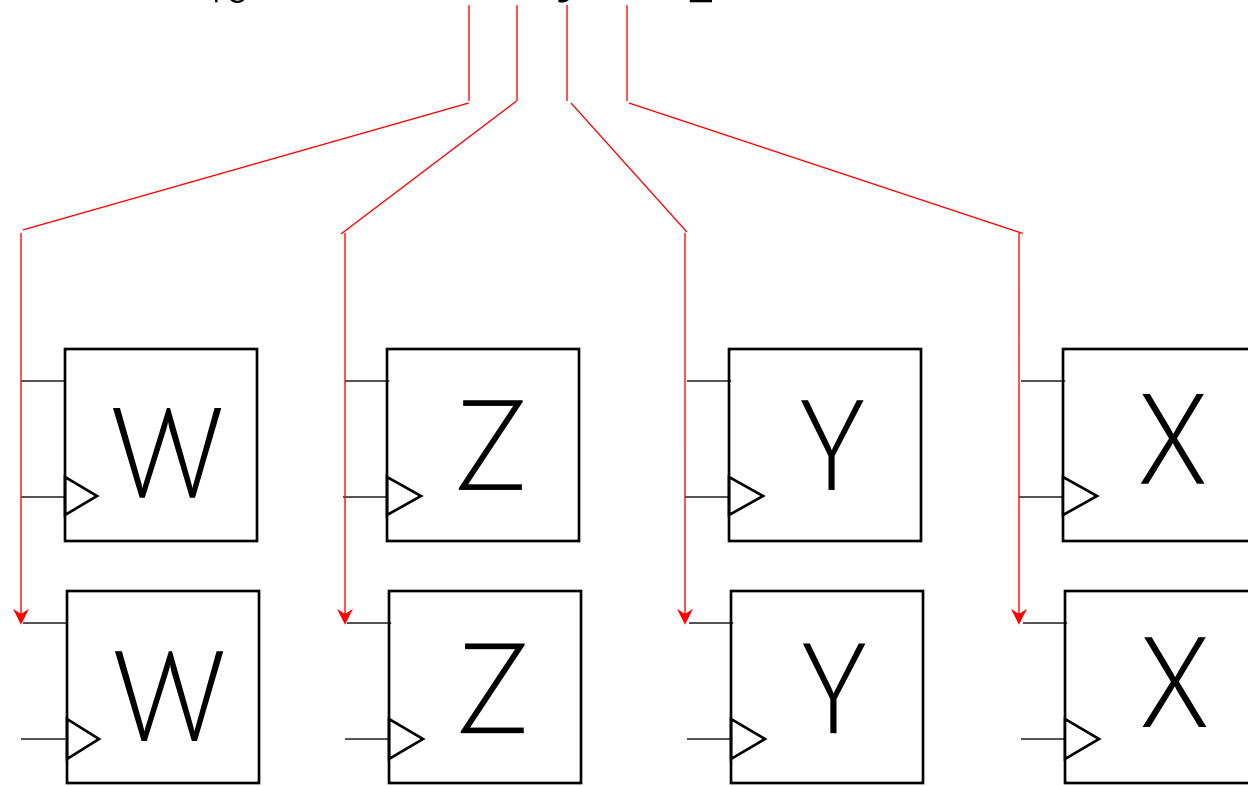
$n=4 \rightarrow$ Each Number $\in [0, 2^4=16) \rightarrow 4$ Cells

$$(N)_{10} = (wzyx)_2$$



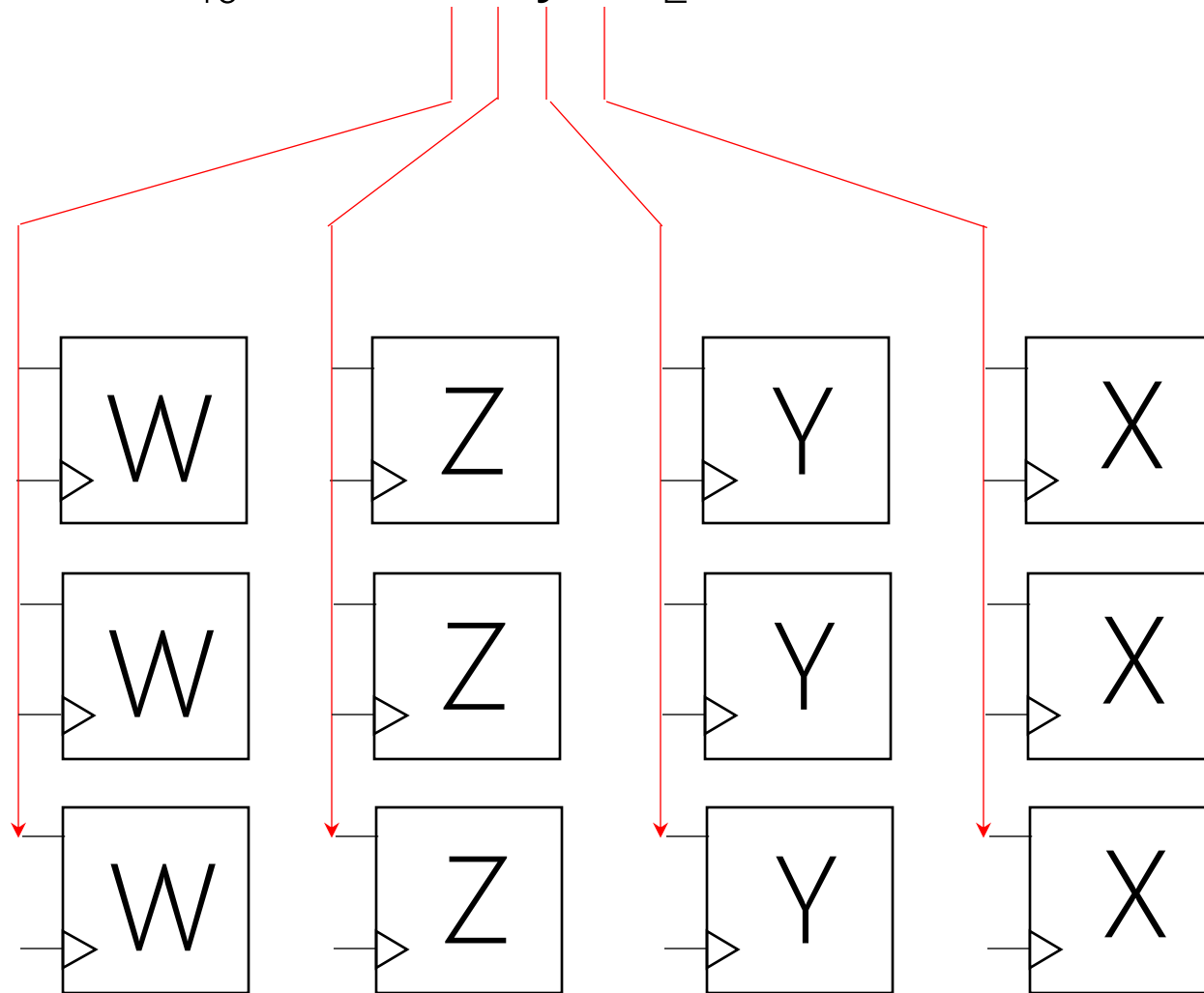
$1 \times \text{Number} \rightarrow 1 \times n \text{ Cells} \rightarrow 1 \times n\text{-bit Register}$
 $\text{Register} \rightarrow \text{a vector of Cells}$

$$(N)_{10} = (wzyx)_2$$



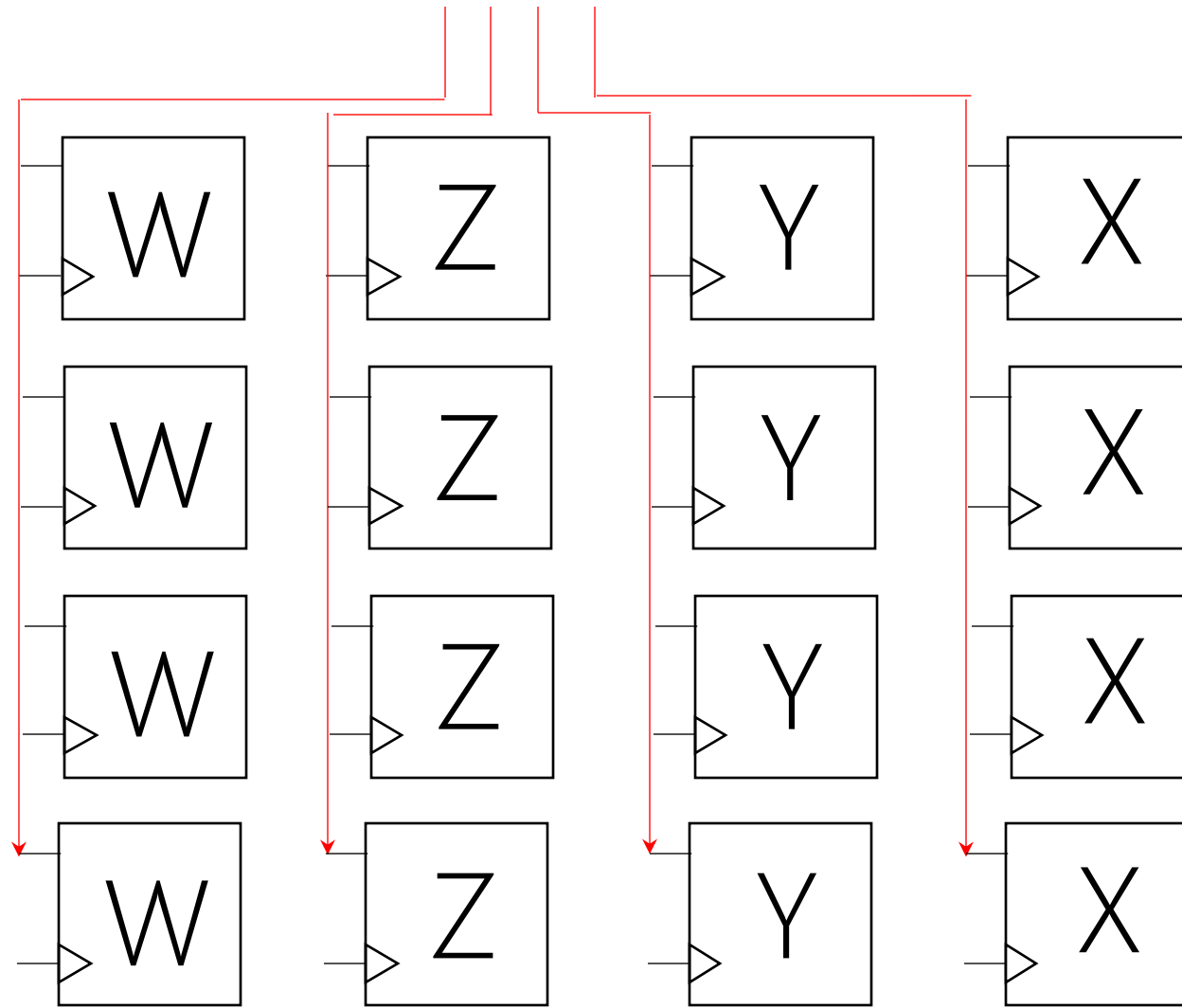
2 × Numbers → 2 × n Cells → 2 × n-bit Registers

$$(N)_{10} = (wzyx)_2$$



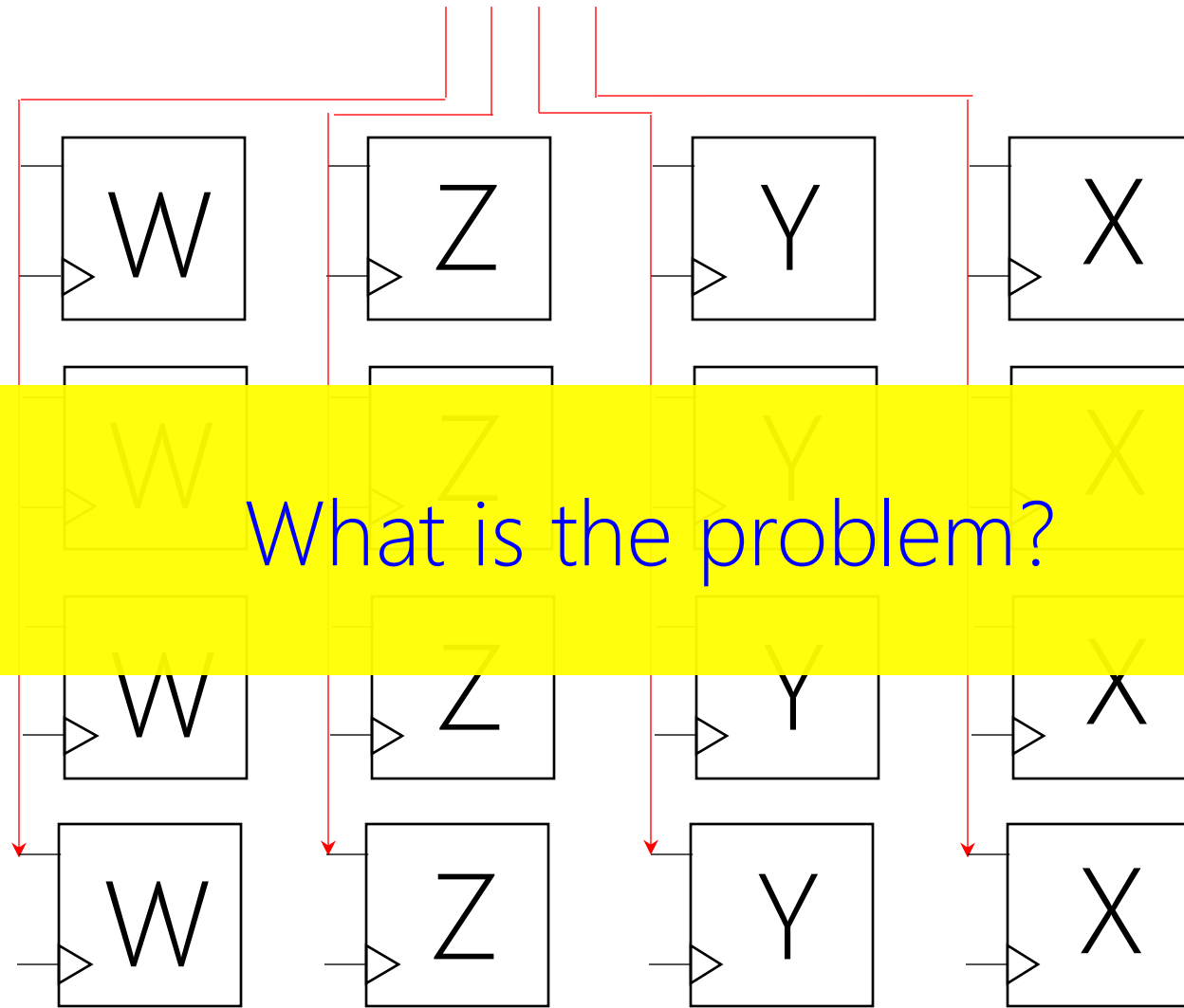
3 × Numbers → 3 × n Cells → 3 × n-bit Registers

4-bit Data Bus



4 × Numbers → 4 × n Cells → 4 × n-bit Registers

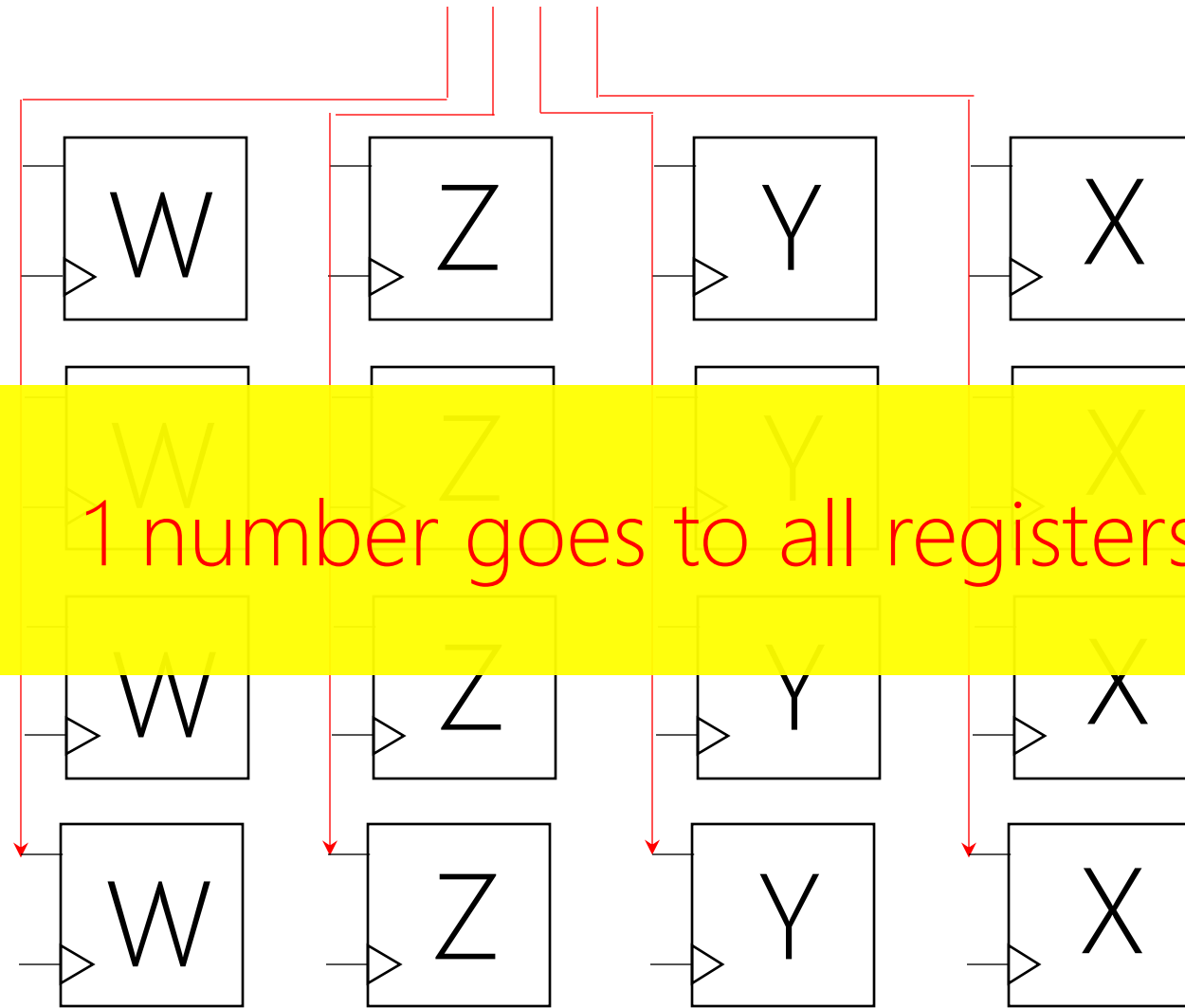
4-bit Data Bus



What is the problem?

4 × Numbers → 4 × n Cells → 4 × n-bit Registers

4-bit Data Bus

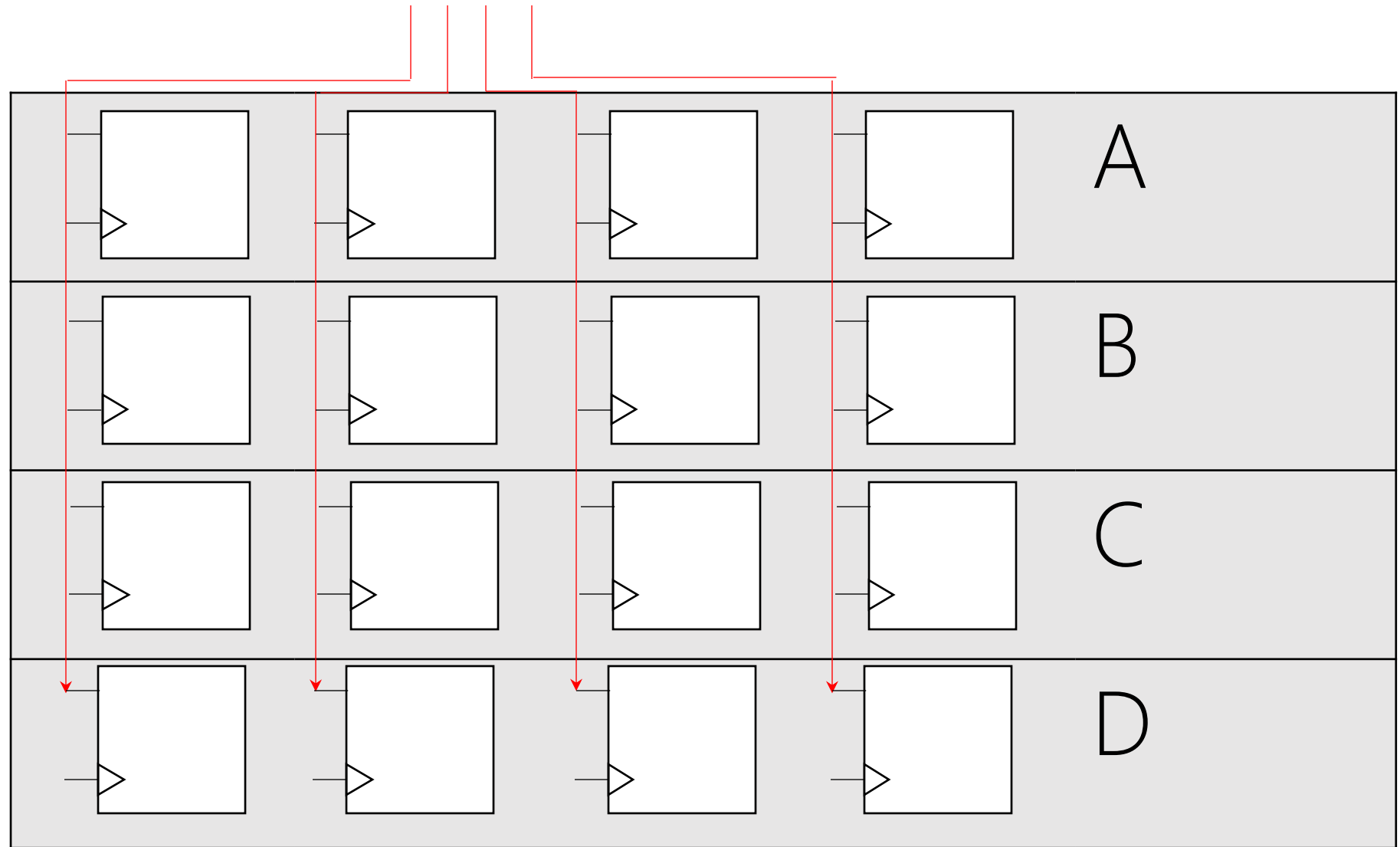


$4 \times \text{Numbers} \rightarrow 4 \times n \text{ Cells} \rightarrow 4 \times n\text{-bit Registers}$

Which Register?

Select Register by Name?
If there are few rows

4-bit Data Bus



4 × Numbers → 4 × n Cells → 4 × n-bit Registers

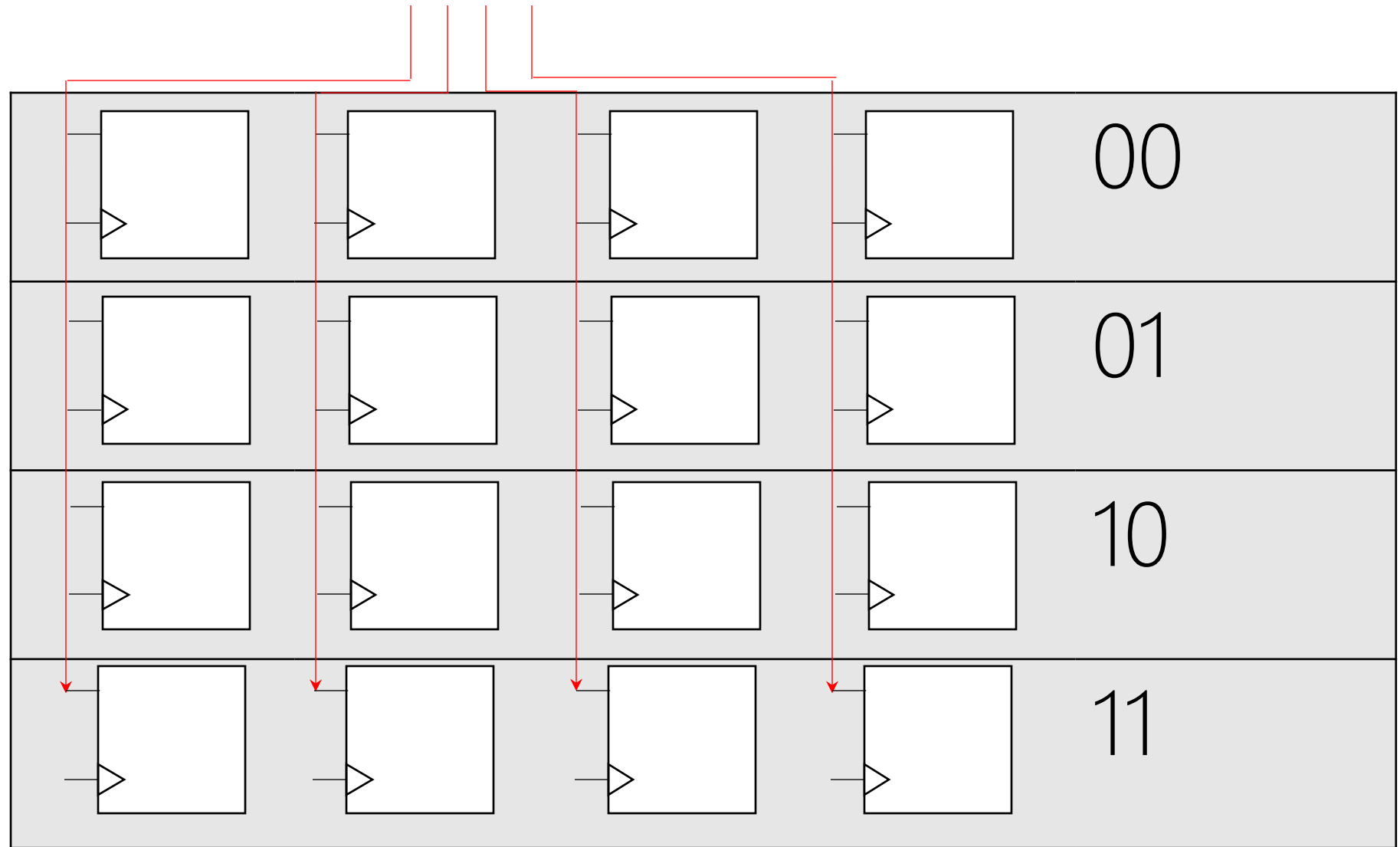
Select Register by Name?

If there are few rows

But what if we have thousands of rows?

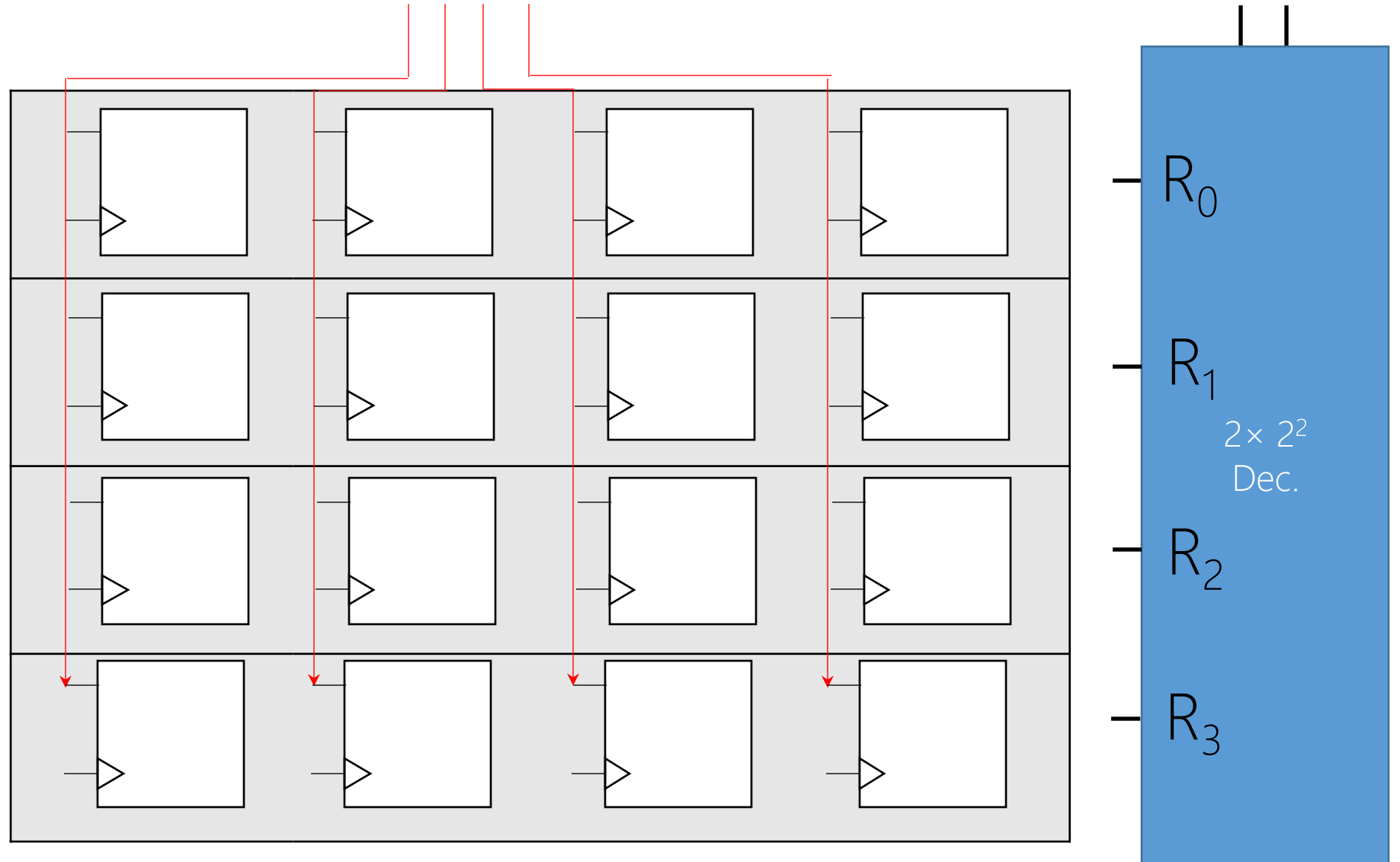
Select Register by Address?

4-bit Data Bus



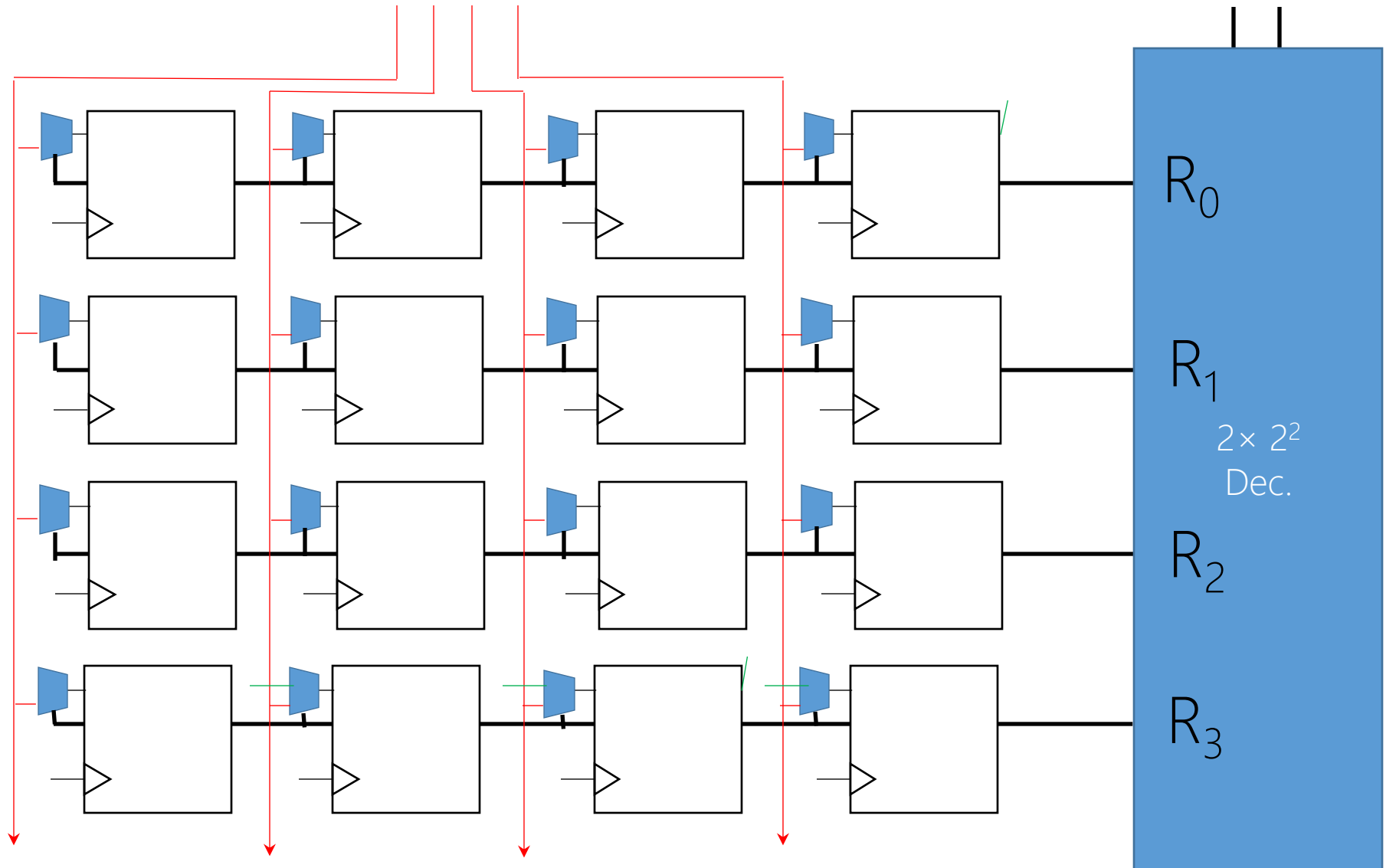
4-bit Data Bus

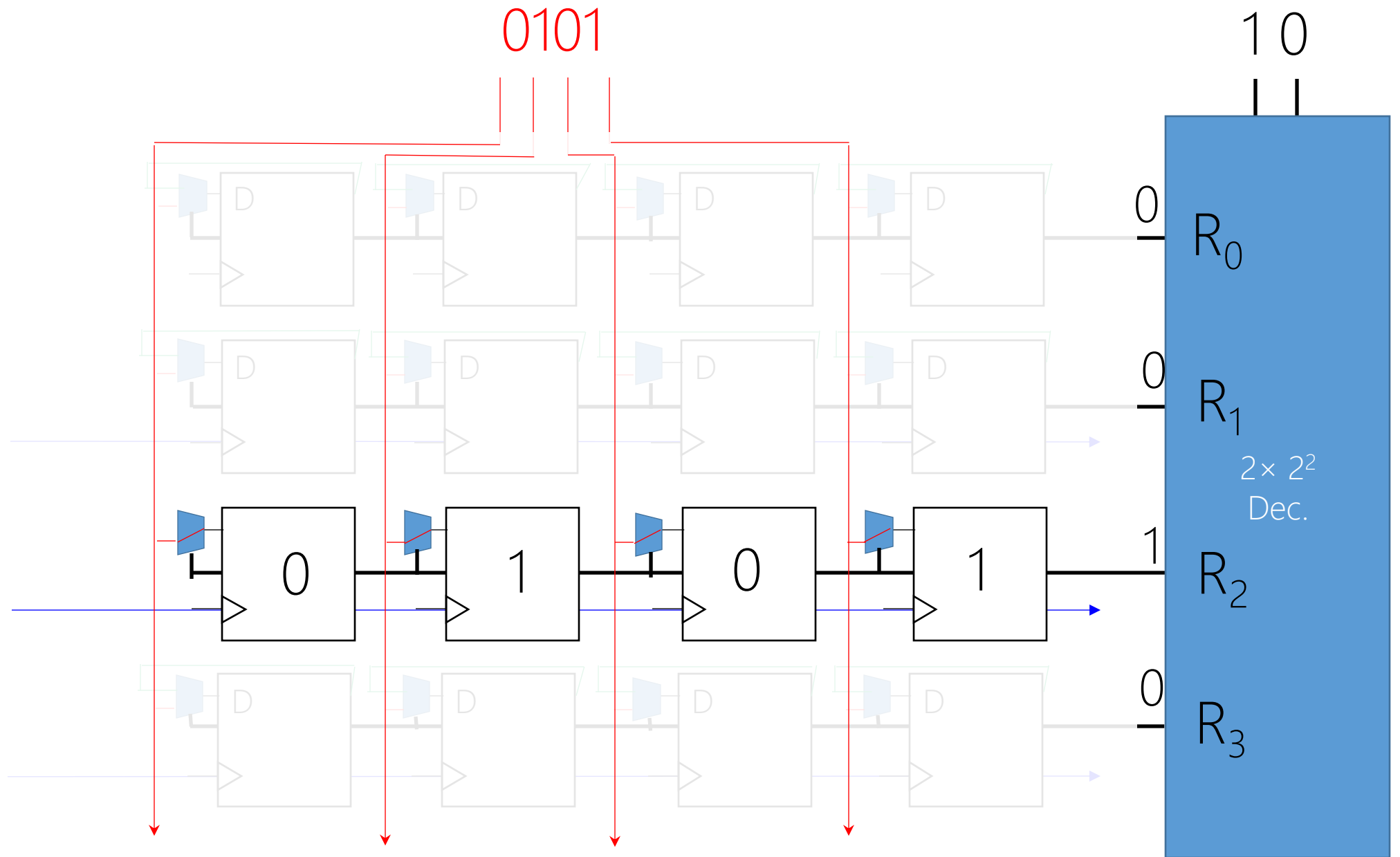
2-bit Address Bus

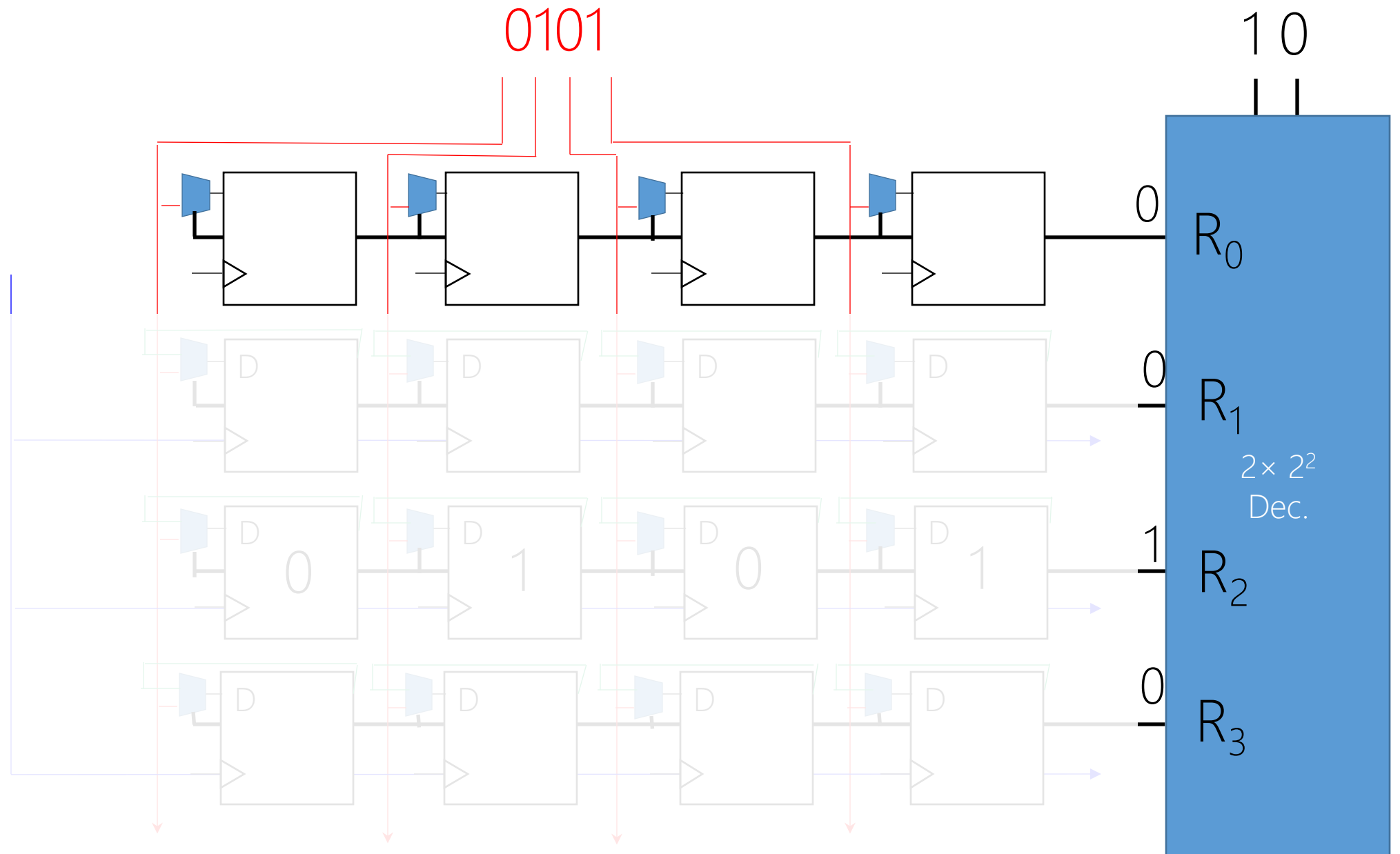


4-bit Data Bus

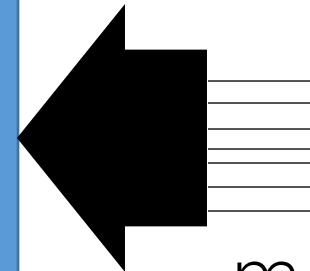
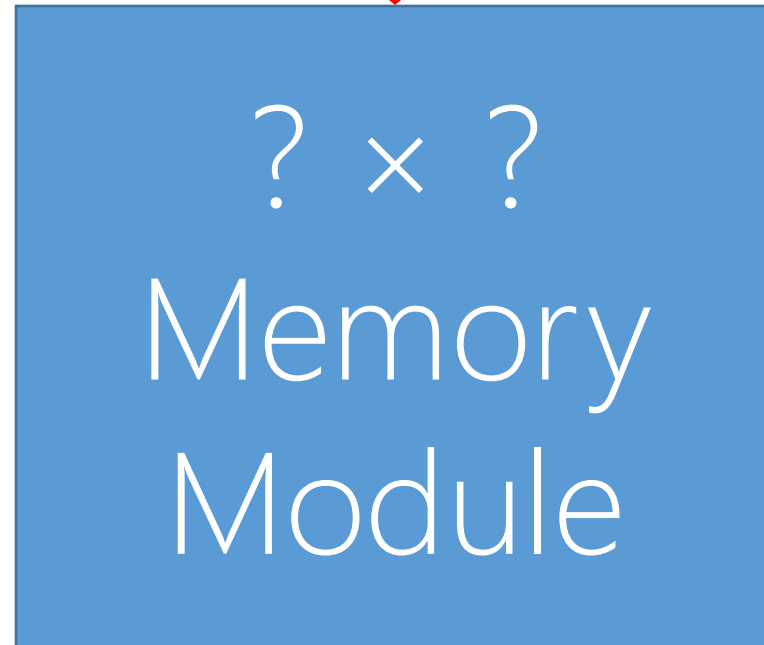
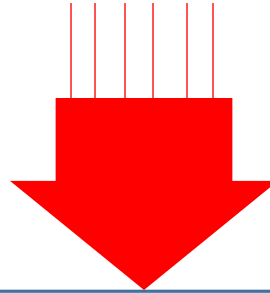
2-bit Address Bus





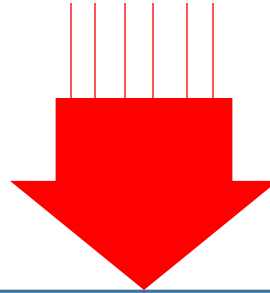


n-bit Data Bus

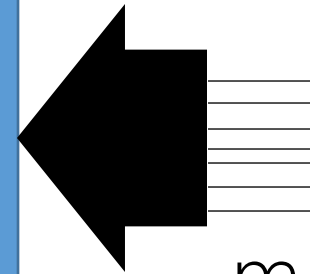


m-bit Address Bus

n-bit Data Bus



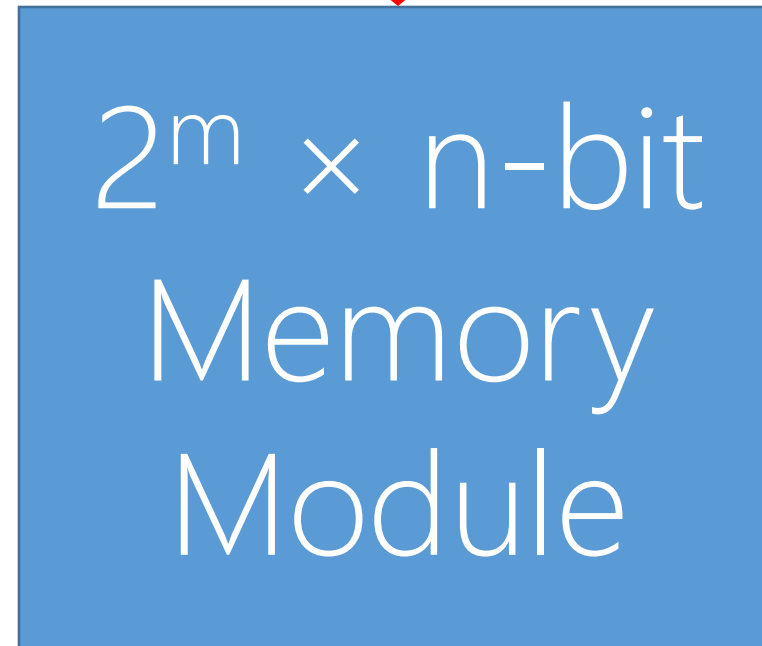
$2^m \times n\text{-bit}$
Memory
Module



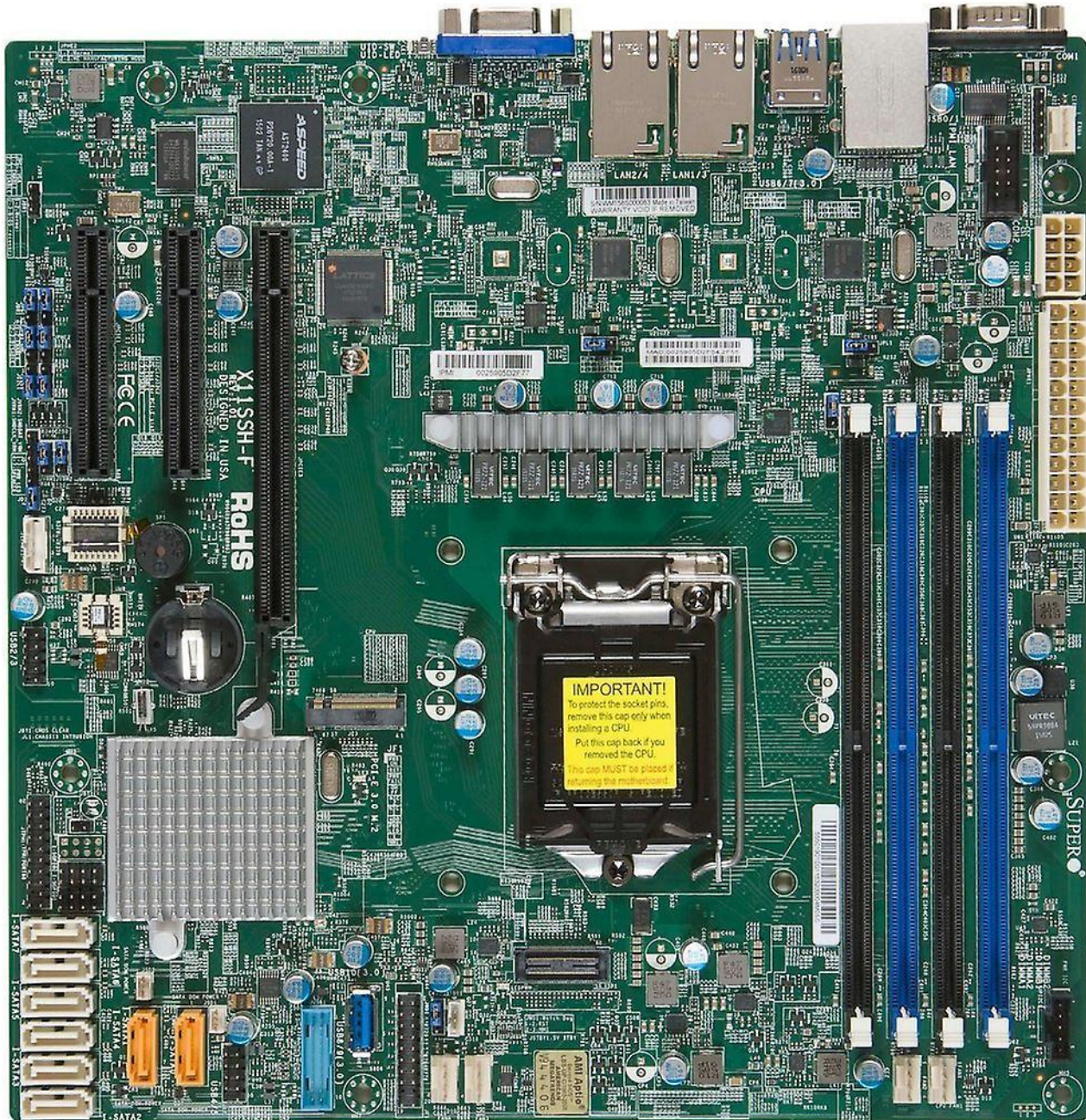
m-bit Address Bus

n-bit Data Bus

Size of bus is fixed!



m-bit Address Bus



64-bit Data Bus:

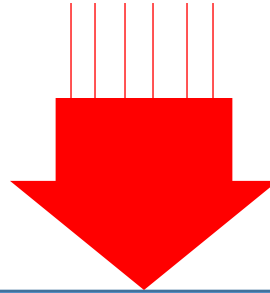
Numbers $\in [0, 2^{64})$

Long Unsigned Integer in C/C++
How about signed integers?

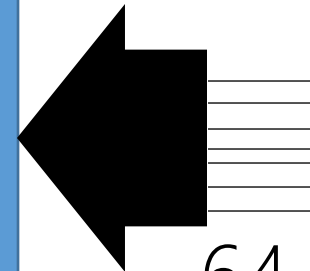
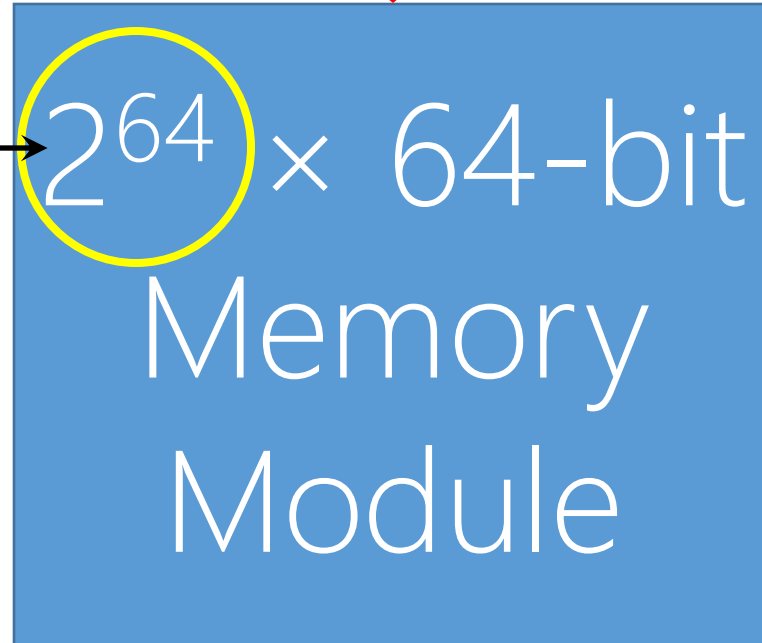
64-bit Address Bus:

2^{64} Memory Locations

64-bit Data Bus



Buying 2^{64} !
You must be rich.
You don't need it



64-bit Address Bus

$2^{64} = 2^{10} \times 2^{10} \times 2^{10} \times 2^{10} \times 2^{10} \times 2^{10} \times 2^4 = 16 \text{ Exa Memory Locations}$

Kilo → Mega → Giga → Peta → Exa

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$16\text{ G} = 16 \times 2^{10} \text{ (M)} = 16 \times 2^{10} \times 2^{10} \text{ (K)} = 16 \times 2^{10} \times 2^{10} \times 2^{10} \text{ Memory Locations}$



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2666 MT/s

2400 MT/s

3200 MT/s

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$16\text{ G} = 16 \times 2^{10} \text{ (M)} = 16 \times 2^{10} \times 2^{10} \text{ (K)} = 16 \times 2^{10} \times 2^{10} \times 2^{10}$ Memory Locations
How many Cells?



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Size :

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$16\text{ G} = 16 \times 2^{10} \text{ (M)} = 16 \times 2^{10} \times 2^{10} \text{ (K)} = 16 \times 2^{10} \times 2^{10} \times 2^{10} \text{ Memory Locations}$

How many Cells?

$$(2^4=16) \times 2^{10} \times 2^{10} \times 2^{10} \times (2^6=64\text{-bit Data Bus}) = 2^{40}$$



New (15) from **CDN\$ 81.00** + FREE Shipping

Size :

16GB Dual Rank

Style: 2400 MT/s

2666 MT/s

2400 MT/s

3200 MT/s

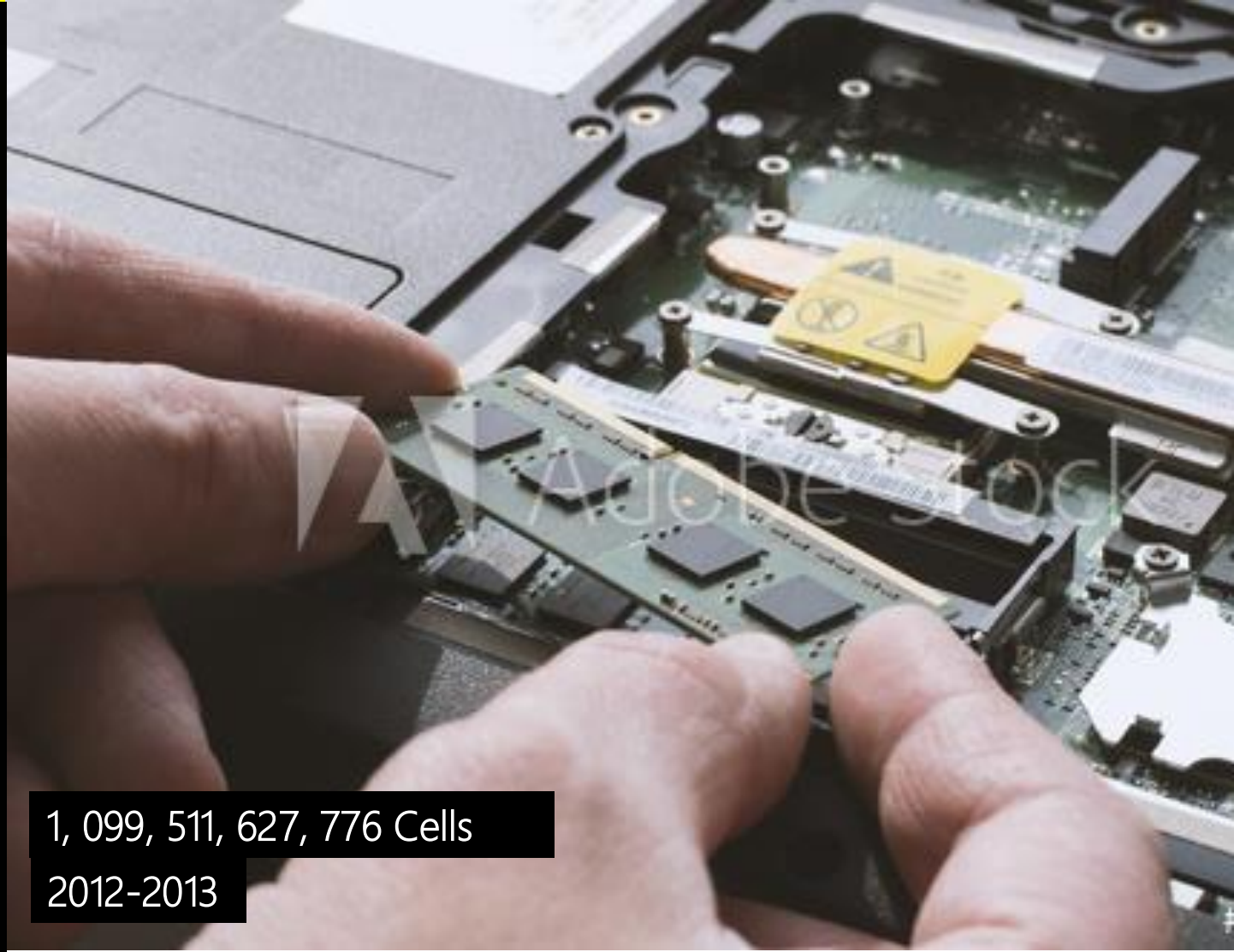
1, 099, 511, 627, 776 Cells

$16\text{ G} = 16 \times 2^{10} \text{ (M)} = 16 \times 2^{10} \times 2^{10} \text{ (K)} = 16 \times 2^{10} \times 2^{10} \times 2^{10} \text{ Memory Locations}$
How many Cells?

$$(2^4=16) \times 2^{10} \times 2^{10} \times 2^{10} \times (2^6=64\text{-bit Data Bus}) = 2^{40}$$

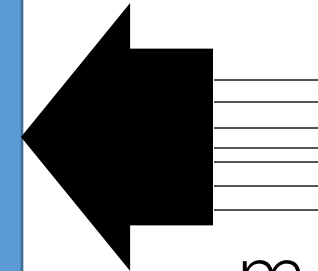
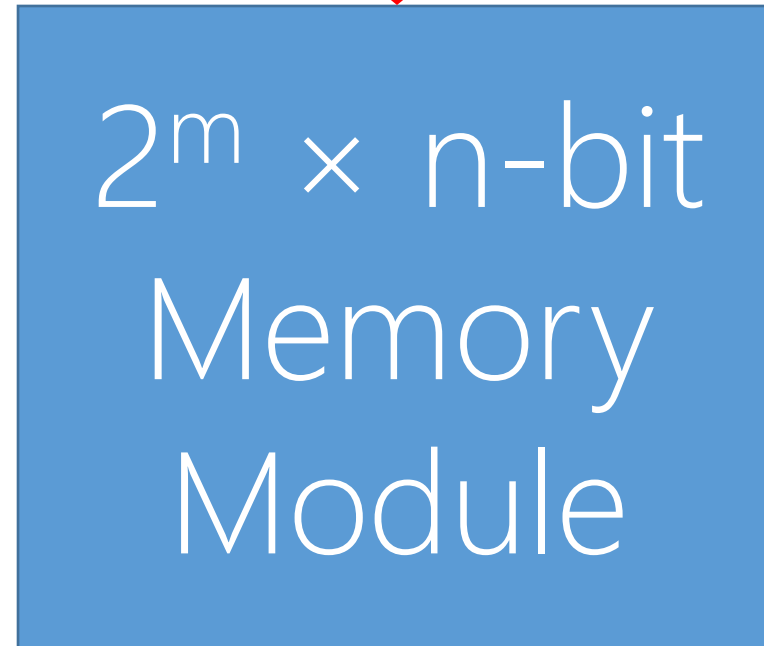
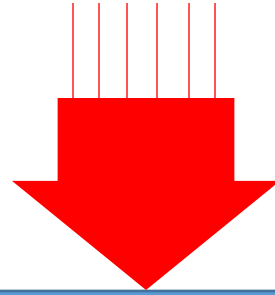


1 Cell
1958–1959

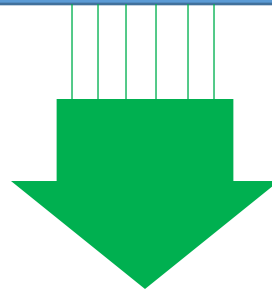


1, 099, 511, 627, 776 Cells
2012–2013

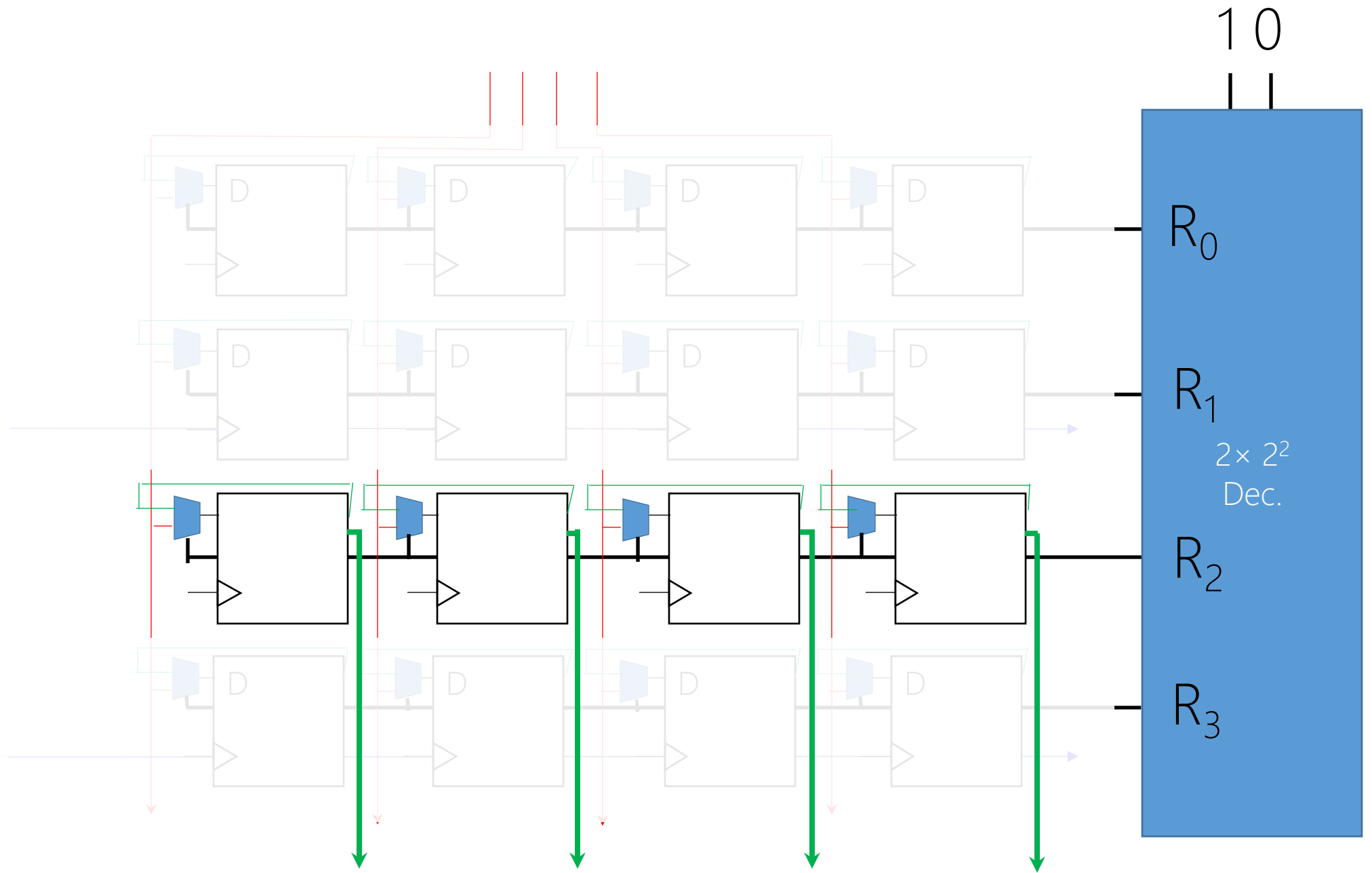
n-bit Data Bus (Write)

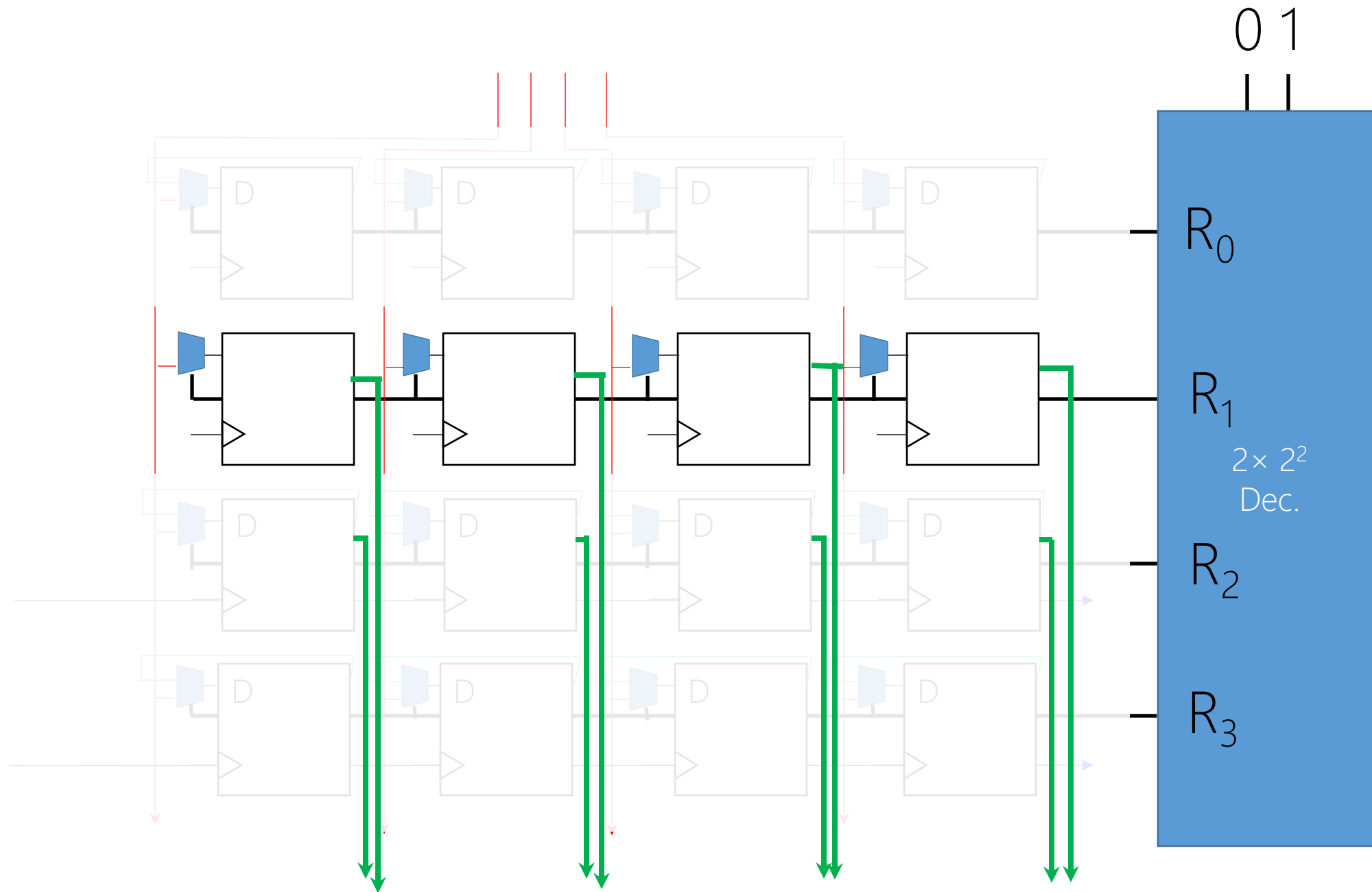


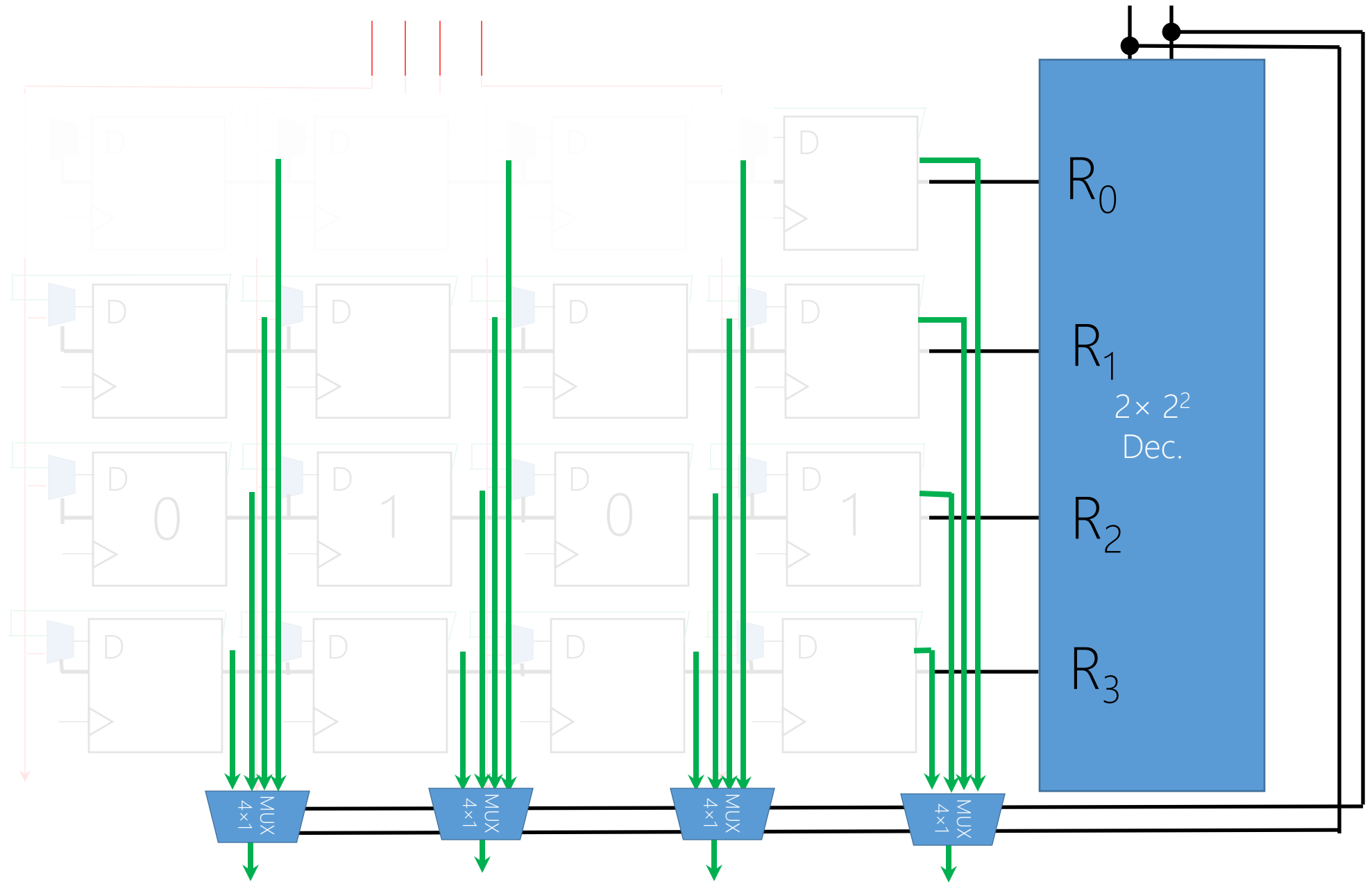
m-bit Address Bus

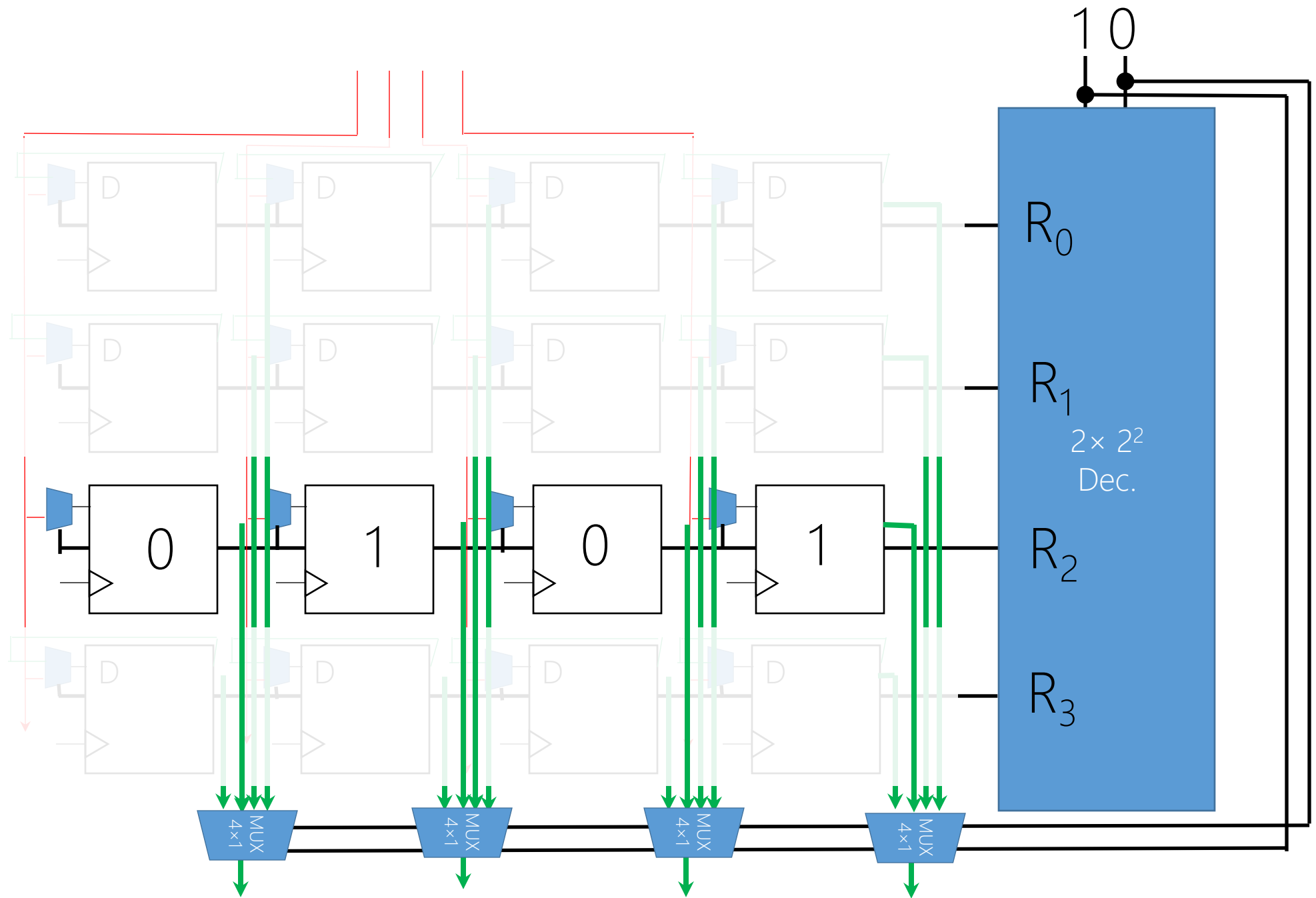


n-bit Data Bus (Read)



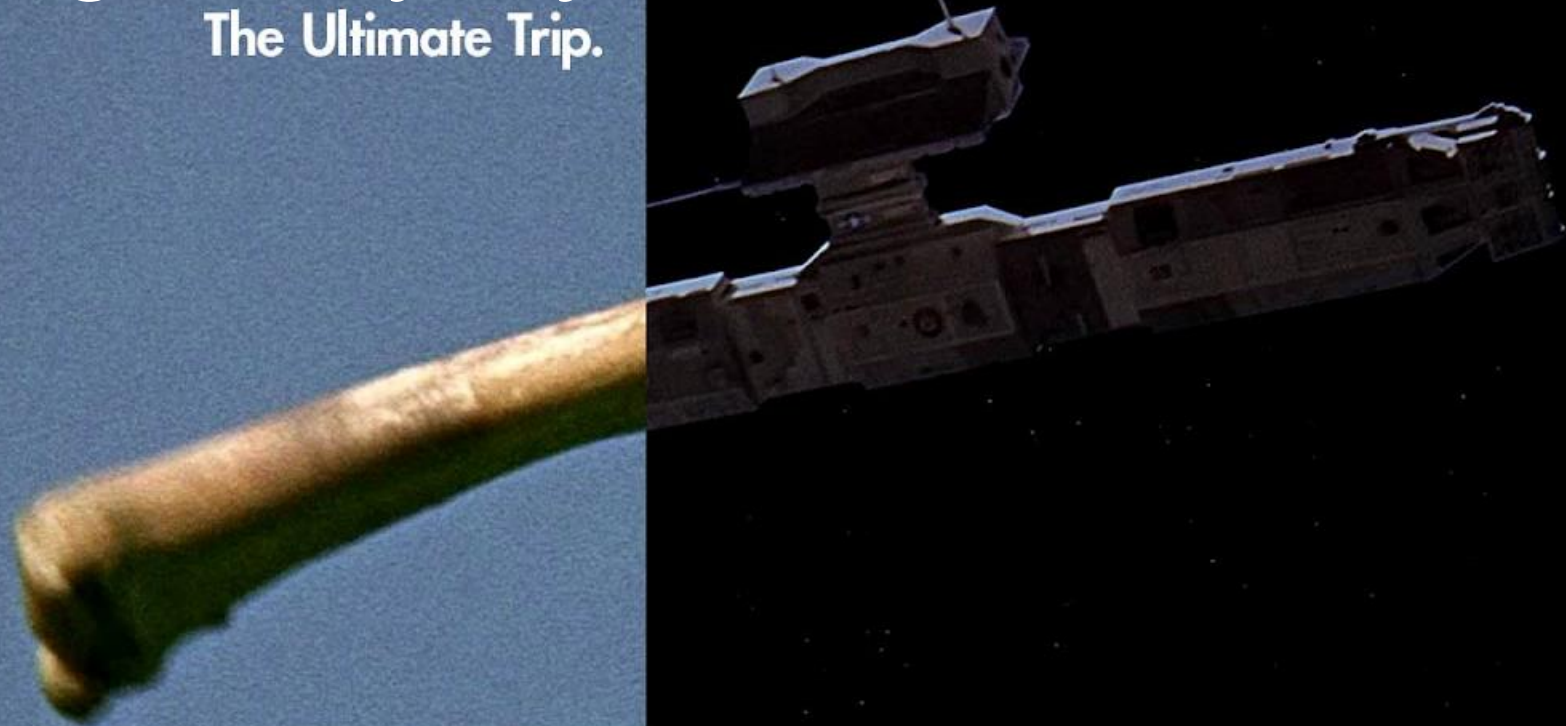






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What is that cell?



1 Cell
1958–1959